
Dynamic Symmetric Point Tracking: Tackling Non-ideal Reference in Analog In-memory Training

Quan Xiao^{*1} Jindan Li^{*1} Zhaoxian Wu¹ Tayfun Gokmen² Tianyi Chen^{1,3}

Abstract

Analog in-memory computing (AIMC) performs computation directly within resistive crossbar arrays, offering an energy-efficient platform to scale large vision and language models. However, non-ideal analog device properties make the training on AIMC devices challenging. In particular, its update asymmetry can induce a systematic drift of weight updates towards a device-specific symmetric point (SP), which typically does not align with the optimum of the training objective. To mitigate this bias, most existing works assume the SP is known and pre-calibrate it to zero before training by setting the reference point as the SP. Nevertheless, calibrating AIMC devices requires costly pulse updates, and residual calibration error can directly degrade training performance. In this work, we present the first theoretical characterization of the pulse complexity of SP calibration and the resulting estimation error. We further propose a dynamic SP estimation method that tracks the SP during model training, and establishes its convergence guarantees. In addition, we develop an enhanced variant based on chopping and filtering techniques from digital signal processing. Numerical experiments demonstrate both the efficiency and effectiveness of the proposed method. Our code will be released at <https://github.com/Jindanli898/E-RIDER>.

1. Introduction

Recent breakthroughs in large vision and language models have been driven by the rapid maturation of modern hardware accelerators, including GPU, TPU (Jouppi et al., 2023),

NPU (Esmaeilzadeh et al., 2012), and emerging AI-specific chips such as NorthPole (Modha et al., 2023). Despite these advances, the energy costs of training and deploying such models still remain prohibitively high (Touvron et al., 2023; Brown et al., 2020).

To mitigate the energy bottleneck, analog in-memory computing (AIMC) has emerged as a promising platform that leverages resistive crossbar arrays to enable in-situ analog matrix-vector multiplications (MVMs). In AIMC, the weight matrix is stored as the conductance of resistive devices on a crossbar array, while MVM inputs and outputs are encoded as analog voltages and currents. By leveraging Kirchhoff’s and Ohm’s laws, AIMC devices perform MVMs directly without data movement, achieving $10\times$ - $10,000\times$ lower inference energy than GPUs (Jain et al., 2019; Cosemans et al., 2019; Papistas et al., 2021).

Consider a model training problem with the objective $f(\cdot) : \mathbb{R}^D \rightarrow \mathbb{R}$ and its model parameters $W \in \mathbb{R}^D$ by

$$W^* := \arg \min_{W \in \mathbb{R}^D} \{f(W) := \mathbb{E}_{\xi}[f(W; \xi)]\} \quad (1)$$

where ξ is a random data sample. Different from digital training, on AIMC hardware, the weights are updated by the so-called *pulse update*. When receiving electrical pulses, a resistive element updates its conductance based on its pulse polarity (Gokmen & Vlasov, 2016). At each pulse cycle, the weight changes by either $\Delta w_{\min} \cdot q_+(w)$ or $-\Delta w_{\min} \cdot q_-(w)$, where $\Delta w_{\min} > 0$ is the known *response granularity*, and $q_+(w)$ and $q_-(w)$ are fixed but unknown *response functions*, that describe how current weight w responds to a small weight change $\Delta w_{\min}$. The response granularity $\Delta w_{\min}$ is one source of training errors, since it quantizes the desired gradients into finite conductance changes. Another key challenge in analog training is *update asymmetry*: at a fixed weight w , the conductance change induced by a positive pulse, $q_+(w)$, generally differs from that induced by a negative pulse, $q_-(w)$. This asymmetry stems from the underlying asymmetric physical mechanisms of analog devices, such as unequal conductive filament formation and rupture in HfO_2 -based ReRAM devices (Jang et al., 2014), or nonreciprocal crystallization and amorphization in PCM devices (Burr et al., 2016). Although asymmetric devices offer practical advantages, such as faster response

^{*}Equal contribution ¹Department of Electrical and Computer Engineering, Cornell University, New York, NY ²IBM T. J. Watson Research Center, Yorktown Heights, NY ³Rensselaer Polytechnic Institute, Troy, NY. Correspondence to: Tianyi Chen <chentianyi19@gmail.com>.

and lower pulse voltage (Stecconi et al., 2024), making them attractive for deployment, their asymmetric update behavior can induce systematic weight drift during analog training.

By decomposing $q_+(w)$ and $q_-(w)$ into a *symmetric component* and an *asymmetric component* that capture the device-specific drift (will explain in (6)), we model the analog update as a scaled desired update plus an asymmetric drift term. Mathematically, each resistive element admits a device-specific *symmetric component* $F(\cdot)$, an *asymmetric component* $G(\cdot)$. To apply an increment $\Delta W_k = \alpha \nabla f(W_k; \xi_k)$ with stepsize α , the analog device performs the following Analog Update (Wu et al., 2025; Li et al., 2025)

$$W_{k+1} = W_k + \Delta W_k \odot F(W_k) - |\Delta W_k| \odot G(W_k) + b_k \quad (2)$$

where $|\cdot|$ and $\odot$ denote the coordinate-wise absolute value and multiplication, and b_k quantifies the stochastic discretization error of sending a finite number of pulses of length $\Delta w_{\min}$ to change each d -th element of W_k . Unlike on digital devices, where $G(\cdot) \equiv 0$, $G(\cdot) \neq 0$ on analog devices induces the *update asymmetry*. Specifically, even without the discretization error b_k , the training optimum W^* is generally not a stationary point of the Analog Update because

$$\mathbb{E}[W_{k+1} | W_k = W^*] = W^* - \alpha \mathbb{E}_\xi[|\nabla f(W^*; \xi)|] \odot G(W^*)$$

does not equal W^* whenever the device-asymmetry term is nonzero (i.e., $G(W^*) \neq 0$) as $\mathbb{E}_\xi[|\nabla f(W^*; \xi)|] > 0$. This motivates us to study the symmetric points where we do not have device-asymmetry as defined below.

Definition 1.1 (Symmetric point). A point is termed *symmetric point* (SP) if the following equation holds

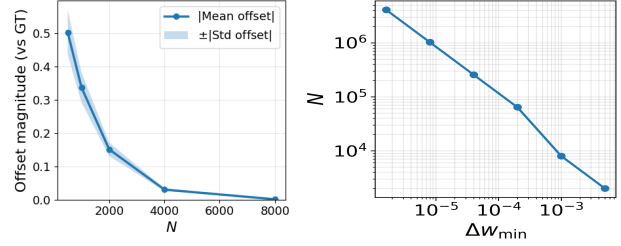
$$G(W^\diamond) = 0. \quad (3)$$

With this, SGD with Analog Update can be interpreted as implicitly minimizing a regularized objective of the form

$$\min_W f(W) + \Theta(\sigma^2 \|W - W^\diamond\|^2) \quad (4)$$

where σ^2 denotes the variance of the stochastic gradient noise (Wu et al., 2025). Consequently, obtaining an accurate estimate of $W^\diamond$ is a key prerequisite for analog training algorithm design, as it enables effective compensation of the drift induced by *update asymmetry*.

In existing algorithm design and convergence theory of analog training (Wu et al., 2024; 2025), $W^\diamond$ is assumed to be 0 for simplicity. However, in practice, the SP is device-specific, so it requires per-device estimation and calibration to zero before training (Gokmen, 2021). To estimate the SP, a typical zero-shifting (ZS) algorithm has been proposed by



(a) Offset vs. pulse budget. (b) Pulse cost v.s. $\Delta w_{\min}$

Figure 1. Trade-off between SP estimation accuracy and pulse cost for ZS algorithm. (a) For each N , we obtain per-cell SP estimates on a 512×512 array, and compute the mean and standard deviation across all cells. We plot the offsets of these statistics relative to the ground truth. (b) As $\Delta w_{\min}$ decreases, achieving a target accuracy (e.g., $\leq 1\%$ relative mean error) needs substantially more pulses.

Kim et al. (2019), which sends alternative up-down pulses to push the device to its SP. However, as shown in Figure 1, its pulse count significantly impacts the precision of SP estimation, and higher-precision devices (smaller $\Delta w_{\min}$) require more pulses to achieve a target accuracy. These observations lead to a critical question regarding the trade-off between pulse overhead and accuracy for SP estimation:

Q1) How to quantify the pulse efficiency of the SP estimation algorithm in terms of the number of pulses?

After obtaining the estimate of SP and calibrating it to zero (through a reference device), we can leverage existing zero-SP-based analog training algorithms for the model training (Wu et al., 2025; 2024; Gokmen, 2021). However, this renders the subsequent training phase vulnerable to estimation error propagated from the SP estimation stage when we do not have enough pulse budget; see Figure 2. Intuitively, the training process can also provide valuable feedback for SP tracking: if $W^\diamond$ is inaccurate, the resulting compensation deviates from the true drift term in (4), which in turn degrades training. Considering the benefits of integrating SP tracking with training, another natural question is

Q2) Can we develop a dynamic SP tracking algorithm during training to reduce the overall pulse complexity?

To address the above two questions, we develop a dynamic SP-tracking algorithm based on multi-sequence update and the filtering theory from digital signal processing.

1.1. Our contributions

We summarize the contributions of this paper as follows:

C1) We provide the first theoretical complexity analysis of the standard ZS algorithm for SP estimation (Kim et al., 2019). Crucially, we prove that the pulse count required for accurate estimation scales with the inverse device update granularity $\mathcal{O}(\Delta w_{\min}^{-1})$. This reveals a

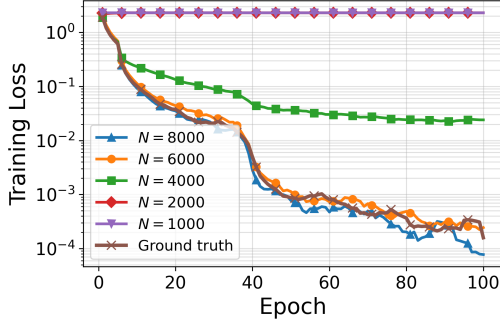


Figure 2. Training loss on MNIST (LeNet-5, TT-v1 (Gokmen & Haensch, 2020)) using ground-truth SP and SPs estimated with different numbers of pulses N via zero-shifting Algorithm 1.

fundamental **device dilemma**: as hardware becomes more precise, static SP estimation becomes more expensive, motivating the need for dynamic tracking.

- C2) We propose a novel Residual learning with Dynamic symmetric point tracking (RIDER) algorithm that jointly performs SP tracking and model training on the analog devices. We proved that the proposed RIDER algorithm can estimate the SP during the model training with reduced pulse complexity, and achieves the same rate $\mathcal{O}(1/\sqrt{K})$ as standard SGD.
- C3) Building on C2), we propose an Enhanced variant of RIDER, termed E-RIDER, that leverages the frequency theory to accelerate the SP estimation and a periodic synchronization mechanism to reduce weight programming costs, making the method feasible for energy-constrained edge implementations.
- C4) Numerical experiments validate the effectiveness and efficiency of the E-RIDER for simultaneous SP tracking and model training. With a more accurate SP estimation, the proposed method is able to improve the analog training accuracy in neural network training, and outperforms other analog training algorithms.

1.2. Related works

AIMC accelerates deep learning by executing compute-intensive MVM operations directly within resistive crossbar arrays (Yao et al., 2017; Wang et al., 2019). However, hardware imperfection introduces errors into the training. A series of works proposes methods to offload parts of error-sensitive operations on digital circuits to mitigate the issues (Nandakumar et al., 2020; Wan et al., 2022; Yao et al., 2020). While effective, this digital overhead compromises the intrinsic energy and latency benefits of the architecture. Consequently, this paper focuses on the more challenging regime of fully on-chip training, which maximizes operations in the analog domain to preserve efficiency.

The primary challenges of on-chip training arise from inherent hardware imperfections, such as asymmetric up-

Algorithm 1 ZS algorithm (Kim et al., 2019)

- 1: **Inputs**: initialization W_0 ; response granularity $\Delta w_{\min}$
- 2: **for** $n = 0, 1, \dots, N - 1$ **do**
- 3: draw $\epsilon_n \in \mathbb{R}^D$ with $\epsilon_n^{[d]} \sim \mathcal{U}(\{-\Delta w_{\min}, \Delta w_{\min}\})$
- 4: update W_{n+1} via analog pulse update, i.e. (7)
- 5: **end for**
- 6: **outputs**: W_N

date (Burr et al., 2015; Chen et al., 2015), reading/writing noise (Agarwal et al., 2016; Zhang et al., 2022; Deaville et al., 2021), device/cycle variations (Lee et al., 2019). Researchers have developed various architectural and algorithmic solutions to mitigate these imperfections. For example, (Gokmen & Haensch, 2020; Gokmen, 2021; Wang et al., 2020; Huang et al., 2020) introduce an auxiliary analog array to suppress update noise; Li et al. (2025) addresses the constraints of limited update granularity through a multi-tile orchestration method. Complementing these empirical methods, recent works have established rigorous theoretical frameworks to model these imperfections, proposing *residual learning* as a principled mechanism to recover training performance (Wu et al., 2024; 2025). However, their theory requires the SP to be exactly zero, and empirically they are not robust to nonzero SP. Most closely related to our work, Rasch et al. (2024) proposed a dynamic SP tracking method AGAD with strong empirical performance. However, their methodology lacks theoretical guarantees even without chopping. As will be demonstrated in this paper, omitting a residual learning mechanism results in inferior performance compared to the E-RIDER in this work.

2. Pulse Complexity of SP Estimation

In this section, we model the ZS algorithm as a discrete-time dynamic using pulse updates, and analyze its convergence as the number of pulses increases.

2.1. Mathematical Model of Zero-shifting Algorithm

We will first introduce the mathematical formulation of Analog Update in (2), and then abstract the update rule for the ZS algorithm (Kim et al., 2019).

For each AIMC device, we stack the response functions $q_-(w)$ and $q_+(w)$ into $Q_+(W)$ and $Q_-(W) : \mathbb{R}^D \rightarrow \mathbb{R}^D$, and express the analog weight updates as

$$W_{k+1}^{[d]} = \begin{cases} W_k^{[d]} + \Delta W_k^{[d]} \cdot Q_+(W_k)^{[d]}, & \Delta W_k^{[d]} \geq 0, \\ W_k^{[d]} + \Delta W_k^{[d]} \cdot Q_-(W_k)^{[d]}, & \Delta W_k^{[d]} < 0 \end{cases} \quad (5)$$

where $\Delta W_k \in \mathbb{R}^D$ is the target update increment and $A^{[d]}$ denotes the d -th element of $A \in \mathbb{R}^D$. Following (Wu et al., 2025), the symmetric component $F(\cdot)$ and the asymmetric component $G(\cdot)$ of this device are given by

$$F(W) := (Q_-(W) + Q_+(W))/2, \quad (6a)$$

$$G(W) := (Q_-(W) - Q_+(W))/2, \quad (6b)$$

plugging which into (5) leads to the Analog Update (2).

ZS algorithm (Kim et al., 2019) estimates the SP of an analog device by applying positive (up) and negative (down) update pulses alternatively until the weight no longer drifts, which we model as the following stochastic dynamic

$$W_{n+1} = W_n + \epsilon_n \odot F(W_n) - |\epsilon_n| \odot G(W_n) \quad (7)$$

where $\epsilon_n \in \mathbb{R}^D$ and $\epsilon_n^{[d]} \sim \mathcal{U}(\{-\Delta w_{\min}, \Delta w_{\min}\})$ is uniformly randomly generated as either up or down pulse, which is summarized in Algorithm 1. Note that the original ZS algorithm proposed in (Kim et al., 2019) is a cyclic sampling version of Algorithm 1; i.e. $\epsilon_{2n'}^{[d]} = \Delta w_{\min}$ and $\epsilon_{2n'+1}^{[d]} = -\Delta w_{\min}$. As the convergence analysis for the cyclic version is built upon the stochastic version (7) and the convergence rate order of them are the same, we only present the stochastic case in the main paper and defer the cyclic-case result to the Appendix C.3–C.4.

2.2. Convergence Analysis for ZS Algorithm

Following (Wu et al., 2025), we focus on the analog devices with the following training-friendly response functions.

Definition 2.1 (Training-friendly response functions). Response functions $q_+(\cdot)$ and $q_-(\cdot)$ are said to be training-friendly if they satisfy

- **(Positive-definiteness)** there exist $q_{\min} > 0$ and $q_{\max} > 0$ such that $q_{\min} \leq q_+(w) \leq q_{\max}$ and $q_{\min} \leq q_-(w) \leq q_{\max}, \forall w$; and,
- **(Differentiable)** Both $q_+(\cdot)$ and $q_-(\cdot)$ are differentiable.

Training-friendly response functions capture a wide range of AIMC devices: the positive lower bound $q_{\min}$ excludes dead-update regions, the upper bound $q_{\max}$ rules out unbounded gains, and the differentiability is motivated by continuous conductance mechanisms (ion migration, phase boundary movement, electrochemical reactions) in PCM (Burr et al., 2016; Le Gallo & Sebastian, 2020), ReRAM (Jang et al., 2014; 2015; Stecconi et al., 2024), ECRAM (Tang et al., 2018; Onen et al., 2022).

For this family of response functions, the convergence of ZS algorithm in Algorithm 1 can be characterized below.

Theorem 2.2 (Convergence rate of Algorithm 1). *Considering the response functions in Definition 2.1, then the iterates given by Algorithm 1 with N pulses satisfy*

$$\frac{1}{N} \sum_{n=0}^{N-1} \mathbb{E} [\|G(W_n)\|^2] \leq \mathcal{O}\left(\frac{1}{N\Delta w_{\min}}\right) + \Theta(\Delta w_{\min}).$$

The proof of Theorem 2.2 is deferred to Appendix C.1. Theorem 2.2 shows that, for a device with response granularity $\Delta w_{\min}$, the minimal achievable SP estimation error is $\delta = \Theta(\Delta w_{\min})$, meaning that higher-precision devices (smaller $\Delta w_{\min}$) can attain more accurate SP estimates.

However, it also reveals a **device dilemma**: to achieve the same fixed estimation error $\delta \geq \Theta(\Delta w_{\min})$, the required number of pulses scales inversely with the response granularity $\Delta w_{\min}$, which is $N = \mathcal{O}(\delta^{-1}\Delta w_{\min}^{-1})$, suggesting that high-precision devices with smaller $\Delta w_{\min}$ require more pulses for SP calibration in Algorithm 1. In practice, the response granularity $\Delta w_{\min}$ of AIMC device is made sufficiently small to ensure the gradient conversion precision (Rao et al., 2023; Sharma et al., 2024), so that the computational complexity of Algorithm 1 is relatively high.

Remark 2.3. With additional assumptions on device behavior, we can derive tighter bounds for N w.r.t. target error δ for specific classes of response functions; we defer these results (Theorem C.2) to Appendix C.2. Nonetheless, the number of pulses required to achieve a target SP estimation error still remains inversely proportional to $\Delta w_{\min}$.

Empirical evidence. To illustrate the negative impact of $\Delta w_{\min}$ on the pulse complexity of the ZS algorithm, we test its pulse cost on a ReRAM array device presets from IBM AIHWKit simulator (Rasch et al., 2021), which is essentially analog device with linear response functions; see detailed setup in Appendix F.2. Figure 1a shows the empirical estimated offset of the SP mean and standard deviation when $\Delta w_{\min} = 0.001$, defined as the ground-truth statistic minus the estimated statistic. It can be seen that achieving relative 1% error requires more than 2000 number of pulses in ZS algorithm, which is computationally heavy. In Figure 1b, we report the smallest N such that the relative error of the estimated SP mean is within 1% under different device response granularity $\Delta w_{\min}$. The result shows that higher-precision devices (smaller $\Delta w_{\min}$) require substantially more pulses to reach the same target error δ . Moreover, the relationship of $\Delta w_{\min}$ and N is nearly inversely linear, as predicted by our Theorem 2.2.

To study the effect of SP estimation error on model training, we train a LeNet-5 on MNIST with TT-v1 using SPs estimated from Algorithm 1 with different numbers of pulses and report the results in Figure 2. It shows that a smaller N in Algorithm 1 leads to significant training loss degradation and even failure to converge, which motivates the design of dynamic SP tracking algorithm.

3. Dynamic SP Tracking Algorithm

In this section, we aim to develop a dynamic SP tracking algorithm that leverages the inherent SP-attraction property of AIMC devices, i.e., repeated gradient pulse updates drive

the device state towards its SP. The challenge of algorithm design is deferred to Appendix B.3.

3.1. Basic Design

We first design a basic dynamic SP-tracking algorithm based on Residual Learning in (Wu et al., 2025) that rectifies the update asymmetry in AIMC devices. Residual Learning introduces another AIMC device to learn the asymmetric residual and compensate for it through gradient updates. However, it *assumes an exactly zero SP*, which in turn requires precise zero-shifting calibration prior to training. To enable dynamic SP tracking, we introduce an SP-tracking variable $Q_k \in \mathbb{R}^D$ that approaches the SP over iterations. For each iteration k , we aim to solve the following bilevel problem to correct the asymmetry bias

$$\begin{aligned} \min_{W \in \mathbb{R}^D} \quad & \|P^*(W, Q_k) - Q_k\|^2, \\ \text{s. t. } P^*(W, Q_k) \in \quad & \arg \min_{P \in \mathbb{R}^D} f(W + \gamma(P - Q_k)) \end{aligned} \quad (8)$$

where $P - Q_k$ serves as a zero-shifting (residual) vector. The lower-level problem seeks the optimal compensation vector to correct device-induced errors, while the upper-level problem drives the system toward a zero-shifted state. In short, if we can design a SP-tracking sequence Q_k such that $Q_k \rightarrow W^\diamond$, then the optimal solution for (8) is $P^* = Q^* = W^\diamond$ and $W^* = \arg \min_W f(W)$. We make the following assumptions.

Assumption 3.1 (L -smoothness). The objective $f(W)$ is L -smooth, that is for any $W, W' \in \mathbb{R}^D$, it follows

$$\|\nabla f(W) - \nabla f(W')\| \leq L\|W - W'\|. \quad (9)$$

Assumption 3.2 (Unbiasness and bounded variance). The samples $\{\xi_k\}$ are i.i.d. sampled from a distribution. Moreover, the stochastic gradient is unbiased and has bounded variance, i.e., $\mathbb{E}_{\xi_k}[\nabla f(W_k; \xi_k)] = \nabla f(W_k)$ and $\mathbb{E}_{\xi_k}[\|\nabla f(W_k; \xi_k) - \nabla f(W_k)\|^2] \leq \sigma^2$.

Assumption 3.3 (μ -SC condition). The objective $f(W)$ is strongly convex (SC) with modulus μ .

Assumption 3.4. The stochastic discretization error b_k in (2) satisfies $\mathbb{E}[b_k] = 0$ and $\text{Var}[b_k] = \Theta(\alpha \Delta w_{\min})$.

Assumptions 3.1–3.3 are standard in the stochastic optimization (Bottou et al., 2018), and are all used in the pilot theoretical analysis for analog training (Wu et al., 2025). Assumption 3.4 is used and verified in (Li et al., 2025).

With these assumptions, the lower-level solution of problem (8) is unique, given by

$$P^*(W, Q_k) = Q_k + (W^* - W)/\gamma. \quad (10)$$

Basically, the optimal $P^*(W, Q_k) - Q_k$ tracks the difference of current W_k towards the optimal solution W^* . Besides, the gradient of the bilevel problem (8) with respect to W can be computed via chain rule

$$\nabla_W \|P^*(W, Q_k) - Q_k\|^2 = -2(P^*(W, Q_k) - Q_k)/\gamma.$$

Let us define $\bar{W}_k = W_k + \gamma(P_k - Q_k)$. Inspired by alternating bilevel algorithms (Chen et al., 2021; Ji et al., 2021; Hong et al., 2020; Ghadimi & Wang, 2018; Maclaurin et al., 2015; Franceschi et al., 2017; 2018; Pedregosa, 2016) and putting P_k and Q_k on different analog devices, the update rules using Analog Update in (2) can be written as

$$\begin{aligned} P_{k+1} = P_k - \alpha \nabla f(\bar{W}_k; \xi_k) \odot F_p(P_k) \\ - \alpha |\nabla f(\bar{W}_k; \xi_k)| \odot G_p(P_k) + b_k \end{aligned} \quad (11a)$$

$$\begin{aligned} W_{k+1} = W_k + \beta(P_{k+1} - Q_k) \odot F_w(W_k) \\ - \beta |P_{k+1} - Q_k| \odot G_w(W_k) + b'_k \end{aligned} \quad (11b)$$

where both b_k, b'_k denote the discretization errors, and we denote the corresponding response functions by F_p, G_p for the device of P_k , and by F_w, G_w for the device of W_k .

Design for Q_k sequence. We first note that the SP-drifting issue arises only in the presence of stochastic noise (see (4)), i.e., the device used for P_k , as the W_k sequence is conditionally deterministic given P_{k+1} and Q_k , so we only need to estimate the SP of the device for P_k . Our key observation is that the update of P_k in (11a) combines both descent on $f(\bar{W})$ via a stochastic-gradient increment, and an additional increment that drives $G_p(P_k)$ towards zero, scaled by a positive stepsize $|\nabla f(\bar{W}_k; \xi_k)|$. This suggests that the update of P_k contains an inherent component that pulls it towards the SP. Therefore, we propose a moving averaging sequence Q_k to magnify the SP-attracting property of P_k sequence

$$Q_{k+1} = (1 - \eta)Q_k + \eta P_{k+1} \quad (12)$$

where the Q_k sequence is updated on the digital device so that there is no analog update bias. The complete Residual learning with Dynamic symmetric point tRacking (RIDER) algorithm is summarized in Algorithm 2. The following lemma formalizes the key insight that moving averaging amplifies the SP-attraction property.

Lemma 3.5. Let $W^\diamond$ be the SP used for P_k , i.e. $G_p(W^\diamond) = 0$. For any iteration k , if $\cos(P_{k+1} - W^\diamond, P_{k+1} - Q_k) > 0$, then there exists $\eta \in (0, 1)$ in (12) such that

$$\|Q_{k+1} - W^\diamond\|^2 < \|P_{k+1} - W^\diamond\|^2. \quad (13)$$

The proof of Lemma 3.5 is provided in Appendix D.1. Because the update of P includes a gradient-descent term for the objective in addition to SP drift, so that both $W^\diamond$ and

Algorithm 2 RIDER algorithm

- 1: **Inputs:** initialization P_0, Q_0, W_0 ; residual parameter γ ; response granularity $F_p(\cdot), F_w(\cdot), G_p(\cdot), G_w(\cdot)$
- 2: **for** $k = 0, 1, \dots, K - 1$ **do**
- 3: evaluate $\bar{W}_k = W_k + \gamma(P_k - Q_k)$
- 4: sample stochastic gradient $\nabla f(\bar{W}_k; \xi_k)$
- 5: update P_{k+1} on analog device via (11a)
- 6: update Q_{k+1} on digital device via (12)
- 7: update W_{k+1} on analog device via (11b)
- 8: **end for**
- 9: **outputs:** $\{P_K, Q_K, W_K\}$

Q_k tend to sit on the same side from P_{k+1} . This suggests that the angle condition is likely to hold so that Q_k sequence remains closer to $W^\diamond$ than P_k sequence.

To analyze the convergence of Algorithm 2 with respect to three sequences, we define the convergence metric as

$$E_K = \frac{1}{K} \sum_{k=0}^{K-1} \mathbb{E} [\|W_k - W^*\|^2 + \mathcal{O}(\|P_k - Q_k\|^2) + \mathcal{O}(\|G_p(P_k)\|^2)] \quad (14)$$

where the three terms quantifies the convergence of W_k, Q_k and P_k , respectively. For simplicity, the constants in front of some terms in E_K are hidden. When $E_k \rightarrow 0$, it can be seen that $W_k \rightarrow W^*$ and $P_k, Q_k \rightarrow W^\diamond$. To prove the convergence, we need the following assumption.

Assumption 3.6 (Rayleigh-type lower bound). There exists $C_* > 0$ such that for all k , $\mathbb{E}_{\xi_k} [\|\nabla f(\bar{W}_k; \xi_k)\|_d] \geq C_*$.

Assumption 3.6 is mild since the operations in the analog domain intrinsically involve thermal and electrical noise.

Theorem 3.7 (Convergence of Algorithm 2). Suppose Assumptions 3.1-3.4, 3.6 hold and the response functions F_p, G_p, F_w, G_w satisfy Definition 2.1. If $C_* \geq \frac{4\sqrt{2}\sigma}{\mu} \left(\frac{q_{\max}}{q_{\min}}\right)^{\frac{3}{2}}$, let $\alpha = \Theta\left(\frac{1}{\sqrt{K}}\right)$, $\beta = \Theta(\alpha\gamma\mu)$, $\eta = \Theta(\alpha\mu)$, $\gamma = \Theta(1)$, it holds that

$$E_K \leq \mathcal{O}\left(\frac{\kappa_1 \kappa_2^5}{\sqrt{K}}\right) + \Theta(\Delta w_{\min}). \quad (15)$$

where $\kappa_1 := L/\mu$ and $\kappa_2 := q_{\max}/q_{\min}$ are the condition number of the function and device.

The proof of Theorem 3.7 is provided in Appendix E. Theorem 3.7 shows that, Algorithm 2 can simultaneously track the SP and perform model training, and the minimal achievable training error $\|W_k - W^*\|^2 \leq \delta$ is $\delta = \Theta(\Delta w_{\min})$. Moreover, the convergence rate of Algorithm 2 matches that of Residual Learning (Wu et al., 2025), although we address a more challenging setting with nonzero and unknown SP.

Algorithm 3 Enhanced version of RIDER (E-RIDER)

- 1: **Inputs:** initialization $P_0, \tilde{Q}_0, W_0$ on analog device and $Q_0 = \tilde{Q}_0$ on digital device; residual parameter γ ; response granularity $F_p(\cdot), F_w(\cdot), G_p(\cdot), G_w(\cdot)$; chopper initialization $c_0 = 1$
- 2: **for** $k = 0, 1, \dots, K - 1$ **do**
- 3: draw the chopper variable c_k via (17)
- 4: **if** $\text{sign}(c_k) \neq \text{sign}(c_{k-1})$ **then**
- 5: correct $\tilde{Q}_k = Q_k$ via weight programming
- 6: **end if**
- 7: sample stochastic gradient $\nabla f(\bar{W}_k; \xi_k)$
- 8: update P_{k+1} on analog device via (18a)
- 9: update Q_{k+1} on digital device via (12)
- 10: update W_{k+1} on analog device via (18b)
- 11: **end for**
- 12: **outputs:** $\{P_K, Q_K, W_K\}$

Remark 3.8. As a theoretical baseline, we consider Residual Learning (Wu et al., 2025) paired with SP estimation via the ZS algorithm, which we refer to as two-stage Residual Learning with ZS algorithm. This two-stage method first estimates a static SP $\hat{W}^\diamond$ with ZS and then fixes $Q_k \equiv \hat{W}^\diamond$ during training. We provide its pseudocode in Appendix B.1.

Corollary 3.9 (Overall pulse complexity). To achieve training accuracy $\delta \geq \Theta(\Delta w_{\min})$, RIDER in Algorithm 2 requires $\mathcal{O}(K) = \mathcal{O}(\delta^{-2})$ number of pulses, but the two-stage Residual Learning with ZS algorithm requires $\mathcal{O}(K + N) = \mathcal{O}(\delta^{-2} + \delta^{-1} \Delta w_{\min}^{-1})$ number of pulses.

Corollary 3.9 directly follows from Theorem 3.7 and Theorem 2.2. This corollary suggests that when the target training error $\delta > \Theta(\Delta w_{\min})$, RIDER offers a clear benefit in pulse complexity. In practical high-precision AIMC devices, $\Delta w_{\min}$ is typically engineered to be as small as possible (e.g. $\Delta w_{\min} = 10^{-4}$) to preserve gradient-conversion fidelity (Rao et al., 2023; Sharma et al., 2024), whereas the acceptable training error δ can be substantially larger to enhance generalization. Consequently, RIDER is expected to require fewer pulses than two-stage approaches.

3.2. Enhancement via Chopping and Filtering

To enhance the empirical performance, we will design a variant of Algorithm 2 through chopping and filtering, inspired by (Rasch et al., 2023).

From the digital signal processing perspective, we can treat P_k and Q_k as two time-domain signals, and view the mapping from P_k to Q_k as a difference equation system (Proakis, 2007). Then the following lemma shows that this system can filter out the high-frequency signal components in P_k .

Lemma 3.10. The moving average update (12) defines a stable low-pass filter from P_k to Q_k with the following

Table 1. Test accuracy on LeNet-5 (MNIST) of different methods under different reference mean/std. Best results are highlighted in bold.

Method	Mean \ Std	0.05	0.2	0.3	0.4	0.7	1.0
TT-v2	0	75.19 $\pm$ 1.0	72.62 $\pm$ 0.9	72.39 $\pm$ 1.1	71.50 $\pm$ 0.6	68.55 $\pm$ 0.9	66.96 $\pm$ 1.3
AGAD		90.81 $\pm$ 0.2	90.69 $\pm$ 1.1	91.02 $\pm$ 0.7	90.42 $\pm$ 0.7	90.15 $\pm$ 0.3	88.11 $\pm$ 0.3
E-RIDER		93.75 $\pm$ 0.1	93.71 $\pm$ 0.2	94.15 $\pm$ 0.6	93.26 $\pm$ 1.3	91.67 $\pm$ 0.3	89.02 $\pm$ 0.3
TT-v2	0.2	75.01 $\pm$ 1.4	72.60 $\pm$ 2.1	71.68 $\pm$ 2.2	70.70 $\pm$ 1.1	66.54 $\pm$ 2.9	66.43 $\pm$ 1.3
AGAD		91.14 $\pm$ 0.8	90.66 $\pm$ 0.9	91.61 $\pm$ 0.1	90.61 $\pm$ 0.9	88.59 $\pm$ 0.6	87.73 $\pm$ 1.1
E-RIDER		93.90 $\pm$ 0.6	93.06 $\pm$ 1.0	93.33 $\pm$ 0.8	93.15 $\pm$ 0.5	91.99 $\pm$ 0.1	89.41 $\pm$ 0.8
TT-v2	0.3	73.64 $\pm$ 0.5	72.50 $\pm$ 1.1	72.66 $\pm$ 0.3	70.70 $\pm$ 1.7	65.43 $\pm$ 2.5	66.78 $\pm$ 1.6
AGAD		90.86 $\pm$ 0.4	89.87 $\pm$ 0.4	90.37 $\pm$ 1.2	89.42 $\pm$ 0.8	89.43 $\pm$ 1.2	86.76 $\pm$ 0.4
E-RIDER		92.45 $\pm$ 1.9	92.15 $\pm$ 1.5	92.27 $\pm$ 0.6	91.45 $\pm$ 1.1	89.74 $\pm$ 0.7	90.26 $\pm$ 1.2
TT-v2	0.4	71.71 $\pm$ 1.8	71.89 $\pm$ 3.3	70.83 $\pm$ 3.1	71.01 $\pm$ 2.8	66.63 $\pm$ 1.2	67.08 $\pm$ 1.6
AGAD		89.63 $\pm$ 1.3	89.79 $\pm$ 1.1	89.99 $\pm$ 1.9	90.16 $\pm$ 0.8	87.32 $\pm$ 1.2	86.54 $\pm$ 1.1
E-RIDER		91.72 $\pm$ 0.4	91.76 $\pm$ 0.4	91.29 $\pm$ 1.1	91.54 $\pm$ 1.5	91.17 $\pm$ 0.5	88.02 $\pm$ 0.3

magnitude of the frequency response

$$|H(e^{j\omega})|^2 = \frac{\eta^2}{1 + (1 - \eta)^2 - 2(1 - \eta) \cos \omega}. \quad (16)$$

The proof of this lemma is provided in Appendix D.2. Observing that the magnitude response is maximized at zero frequency ($\omega = 0$) and minimized at the high frequency ($\omega = \pm\pi$), the moving average operator functions as a low-pass filter. Consequently, it effectively attenuates the high-frequency (sign-flipping) components of P_k .

This motivates us to use a chopper variable to further diversify the frequency for the two components in the P_k update and keep the SP drifting part in the low-frequency band. Specifically, let us define the chopper variable c_k that flips the sign with probability (w. p.) $p \in (0, 1)$, i.e.

$$c_{k+1} = \begin{cases} c_k, & \text{w. p. } 1 - p, \\ -c_k, & \text{w. p. } p. \end{cases} \quad (17)$$

With $\bar{W}_k = W_k + \gamma c_k (P_k - Q_k)$, the update (11) becomes

$$P_{k+1} = P_k - \alpha c_k \nabla f(\bar{W}_k; \xi_k) \odot F_p(P_k) - \alpha |c_k \nabla f(\bar{W}_k; \xi_k)| \odot G_p(P_k) + b_k \quad (18a)$$

$$W_{k+1} = W_k + \beta c_k (P_{k+1} - Q_k) \odot F_w(W_k) - \beta |c_k (P_{k+1} - Q_k)| \odot G_w(W_k) + b'_k. \quad (18b)$$

Frequency domain interpretation. In the frequency domain, P_k comprises two distinct components: a high-frequency, sign-flipping term (corresponding to $c_k \nabla f(\bar{W}_k; \xi_k) \odot F_p(P_k)$) and a low-frequency, slowly varying term (corresponding to $|c_k \nabla f(\bar{W}_k; \xi_k)| \odot G_p(P_k)$). The crucial distinction is that the absolute value in the second term eliminates sign changes, causing it to accumulate rather than oscillating. Following Lemma 3.10, applying a moving average to P_k acts as a low-pass filter, which results in the Q_k sequence suppressing the sign-flipping component and retaining the low-frequency signals only; see Figure 3. As

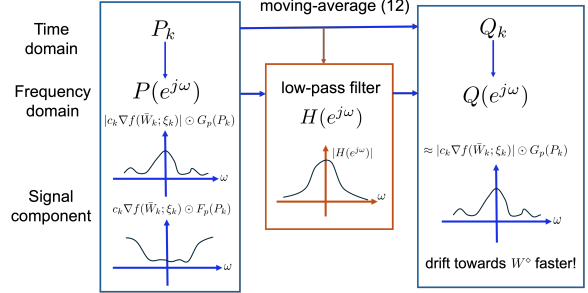


Figure 3. Chopping and filtering via moving average.

the low-frequency component of the P_k update is a scaled increment proportional to $G_p(P_k)$, it forces the Q_k drifting towards the SP of the P device faster.

Furthermore, to stabilize the updates of $\bar{W}_k$, we incorporate the chopper c_k directly into the $\bar{W}_k$ update rule, which is a novel modification compared to prior works (Wu et al., 2025; Rasch et al., 2023). This is because the update term $c_k (P_k - Q_k)$ is stable and (approximately) aligned in sign of $\nabla f(\bar{W}_k)$, which is independent of the flipping. Together, chopping and filtering help the Q_k sequence converge to W^* faster while ensuring the objective descent for $\bar{W}_k$, yielding enhanced empirical performance.

Practical implementation. As Q_k is stored on the digital device, but P_k and W_k are stored on the analog device, computing $P_k - Q_k$ and $P_{k+1} - Q_k$ requires frequent weight programming. To reduce this cost, we store a fake $\tilde{Q}_k$ on an additional analog device and only periodically correct it using the digitally stored Q_k when the sign of c_k flips. Overall, the weight programming cost of the E-RIDER is the same order as the existing dynamic SP tracking method AGAD (Rasch et al., 2023), and the difference of them is summarized in Appendix B.2. We summarize the Enhanced RIDER (E-RIDER) algorithm in Algorithm 3.

4. Experiments

We evaluate our method on MNIST dataset (LeCun et al., 1998b) using a fully analog LeNet-5 (LeCun et al., 1998a)

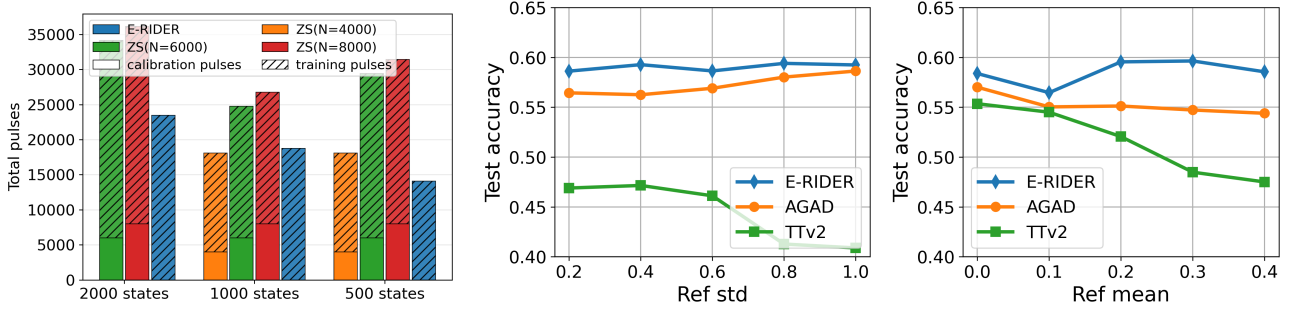


Figure 4. (Left) total pulse cost to reach the target training loss 0.2 on LeNet-5 (MNIST) across different number of states settings. Solid bars indicate the number of pulses using ZS algorithm, while hatched bars indicate the training cost computed as $\text{epochs} \times \lceil \text{data size}/B \rceil \times \text{BL}$, with batch size $B = 64$ and an average update pulse length $\text{BL} = 5$. For 2000 states, ZS ($N = 4000$) fails to reach the target loss. (Middle & right) training loss of E-RIDER and baselines under different reference std/mean on ResNet-18 (CIFAR-100) after 80 epochs.

and a fully analog fully connected network, and on CIFAR-100 dataset (Krizhevsky et al., 2009) by training a ResNet-18 model (He et al., 2016) with the fully connected layer and the last residual block implemented in analog. All experiments are implemented in the AIHWKit simulator (Rasch et al., 2021). Note that RIDER is a special case of E-RIDER with $p = 0$. As shown in the Figure 5, using a small $p > 0$ yields a clear improvement in test accuracy, so we use E-RIDER with the best-tuned p in all subsequent experiments. For fairness, we also tune the chopper probability $p > 0$ for the dynamic SP tracking baseline AGAD (Rasch et al., 2023). Larger-scale experiments of analog ImageNet-1K fine-tuning (Deng et al., 2009), hyperparameters choices for different methods and ablation studies for key hyperparameters for E-RIDER are provided in Appendix F.3–F.6.

Device model. We use two ReRAM array device presets from IBM AIHWKit simulator (Rasch et al., 2021): the HfO_2 -based RRAM preset and the ReRAMArrayOM preset in (Gong et al., 2022b). Both presets inherit from the SoftBoundsReferenceDevice model, where the ReRAM update response is modeled by state-dependent linear soft-bound functions. Let $-\tau_{\min}$ and $\tau_{\max}$ denote the lower and upper weight bounds, respectively, with $\tau_{\min}, \tau_{\max} > 0$. For a scalar weight $w \in [-\tau_{\min}, \tau_{\max}]$, we write the normalized potentiation and depression response functions as

$$q_+(w) = \alpha_+ \left(1 - \frac{w}{\tau_{\max}} \right), \quad q_-(w) = \alpha_- \left(1 + \frac{w}{\tau_{\min}} \right),$$

where $\alpha_+, \alpha_- > 0$ determine the effective update magnitudes for potentiation and depression, respectively. The specific device parameters are provided in Appendix F.1.

E-RIDER achieves better overall pulse complexity. We conduct an ablation study on LeNet-5 (MNIST) with a convergence criterion of training loss ≤ 0.2 . Figure 4 (left) reports the total pulse cost required to reach this target for E-RIDER and the two-stage TT-v2 (Gokmen, 2021) with ZS algorithm under different numbers of device states. For

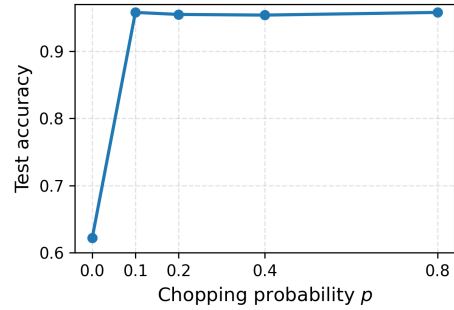


Figure 5. Test accuracy of E-RIDER on MNIST-FCN after 50 epochs under different chopper probabilities p .

the two-stage ZS approach, the total pulse cost is the sum of N pulses for SP estimation and the subsequent training pulses, determined by the number of training epochs K when training with TT-v2 after calibration. When the number of states is low ($\Delta w_{\min}$ is large), SP estimation with fewer pulses (e.g., $N = 4000$) leads to better convergence than using larger N , which consists with Theorem 2.2. However, as the number of states increases (smaller $\Delta w_{\min}$), the two-stage ZS approach becomes more expensive because accurate SP estimation requires a larger pulse budget; for 2000 states, ZS with $N = 4000$ pulses fails to meet the training target loss. Meanwhile, E-RIDER consistently achieves a lower total pulse cost across all settings.

E-RIDER is robust to nonzero SP. We conduct an ablation study on the robustness of E-RIDER to a nonzero SP reference. The device model follows the RRAM-RfO₂ preset (Gong et al., 2022a) to emulate practical update non-idealities in filamentary RRAM. We focus on a limited-state scenario where the number of states is ~ 4 –5. It also mimics the strong device-to-device mismatch and cycle-to-cycle writing noise specified in Appendix F.1. We initialize W° by sampling each entry W_{ij} i.i.d. from a Gaussian distribution with varying reference mean (Ref Mean) and standard deviation (Ref Std) to model different nonzero SP scenarios. We compare E-RIDER with both TT-v2 and AGAD, summarizing the test accuracy after 40 epochs in Tables 1

Table 2. Test accuracy on FCN (MNIST) for different algorithms under different reference mean/std. Best results are highlighted in bold.

Method	Mean \ Std	0.05	0.2	0.3	0.4	0.7	1.0
TT-v2	0	90.26 $\pm$ 0.4	88.75 $\pm$ 0.6	87.40 $\pm$ 0.7	85.98 $\pm$ 0.2	82.85 $\pm$ 2.4	79.20 $\pm$ 3.2
AGAD		92.59 $\pm$ 0.2	91.96 $\pm$ 0.8	92.11 $\pm$ 0.9	92.48 $\pm$ 0.1	92.19 $\pm$ 0.4	91.37 $\pm$ 0.4
E-RIDER		95.45 $\pm$ 0.2	95.48 $\pm$ 0.1	95.39 $\pm$ 0.1	95.34 $\pm$ 0.4	95.42 $\pm$ 0.3	93.86 $\pm$ 0.2
TT-v2	0.2	73.37 $\pm$ 0.3	73.99 $\pm$ 0.7	71.88 $\pm$ 0.4	70.40 $\pm$ 0.8	68.63 $\pm$ 1.1	62.35 $\pm$ 0.6
AGAD		92.86 $\pm$ 1.3	91.95 $\pm$ 0.4	92.63 $\pm$ 0.7	92.80 $\pm$ 0.2	92.22 $\pm$ 0.3	90.77 $\pm$ 0.3
E-RIDER		95.78 $\pm$ 0.1	95.63 $\pm$ 0.1	95.84 $\pm$ 0.3	95.82 $\pm$ 0.3	95.82 $\pm$ 0.3	95.18 $\pm$ 0.3
TT-v2	0.3	72.86 $\pm$ 1.3	72.65 $\pm$ 0.5	69.70 $\pm$ 1.6	69.96 $\pm$ 1.5	66.44 $\pm$ 2.0	61.45 $\pm$ 1.0
AGAD		92.31 $\pm$ 0.4	92.52 $\pm$ 0.4	92.47 $\pm$ 0.5	92.88 $\pm$ 0.3	91.68 $\pm$ 0.5	91.35 $\pm$ 0.4
E-RIDER		95.76 $\pm$ 0.1	95.81 $\pm$ 0.5	95.91 $\pm$ 0.6	95.88 $\pm$ 0.1	95.79 $\pm$ 0.2	94.84 $\pm$ 0.2
TT-v2	0.4	72.00 $\pm$ 2.8	68.08 $\pm$ 2.1	68.80 $\pm$ 1.9	67.85 $\pm$ 1.4	66.15 $\pm$ 0.9	57.79 $\pm$ 0.8
AGAD		92.55 $\pm$ 0.2	92.42 $\pm$ 0.6	92.42 $\pm$ 0.8	92.06 $\pm$ 0.4	90.62 $\pm$ 0.3	90.43 $\pm$ 0.3
E-RIDER		96.07 $\pm$ 0.2	96.21 $\pm$ 0.1	96.20 $\pm$ 0.1	96.23 $\pm$ 0.3	95.87 $\pm$ 0.3	95.14 $\pm$ 0.3

and 2. Each setting is repeated three times, and we report the sample mean and standard deviation results; the hyperparameters are both tuned to optimum (See Appendix F.3 for details). The results demonstrate that: 1) TT-v2 exhibits a substantial accuracy gap to both AGAD and E-RIDER, and its performance degrades markedly as the reference mean/std offsets increase. This is expected because TT-v2 cannot compensate for nonzero reference offsets. 2) E-RIDER consistently outperforms TT-v2 and AGAD in all reference mean/std offset settings, especially with larger SP offset. This is because E-RIDER dynamically tracks SP, and further improves training under low-state devices, which increases the effective weight resolution and enables more accurate gradient updates.

We further evaluate the robustness of our method on CIFAR-100 using a ResNet-18 trained for 80 epochs. We first fix the reference mean to 0.4 and sweep the reference standard deviation, and then fix the reference standard deviation to 0.4 and sweep the reference mean. The results are summarized in Figure 4 (middle & right). We observe similar trends: 1) TT-v2 suffers from a significant degrade in testing accuracy as the reference standard deviation and mean grows, whereas AGAD and E-RIDER remain more stable. 2) E-RIDER consistently achieves higher testing accuracy than both TT-v2 and AGAD, especially for large SP offset.

We additionally provide large-scale ImageNet-1K (Deng et al., 2009) fine-tuning experiments with VGG-11-BN (Simonyan & Zisserman, 2014), a 132M-parameter model with 8 convolutional and 3 fully connected layers, for two dynamic SP-tracking methods, AGAD and E-RIDER, in Appendix F.5. The results further demonstrate the superiority of E-RIDER over AGAD in the large-scale setting.

5. Conclusions

In this paper, we characterize the pulse complexity of ZS algorithm and quantify how the required number of pulse updates for achieving a certain target SP estimation error

scales with device response granularity $\Delta w_{\min}$. Building on these insights, we propose a dynamic SP tracking algorithm, RIDER, which estimates the SP during training, achieving target training accuracy with substantially fewer pulse updates than the two-stage analog training approaches theoretically. We further develop an variant of RIDER, E-RIDER, that accelerates SP tracking through chopper-and-filtering, while reducing weight programming overhead by periodical synchronization. Numerical experiments validate the efficiency and effectiveness of the proposed E-RIDER.

Acknowledgment

The work was supported by the National Science Foundation Projects 2401297 and 2532349, by NVIDIA Academic Grant, by IBM through the IBM-Rensselaer Future of Computing Research Collaboration, and by Cisco Research.

Impact Statement

This paper aims to advance AIMC training via dynamic SP tracking, providing a principled algorithmic framework to mitigate update asymmetry and device drift during learning. By addressing the key challenges in dynamic SP estimation and calibration, our work contributes to the broader development of more robust and hardware-aware training methods for energy-efficient AIMC accelerators. Potential societal impacts include enabling lower-power AI training and adaptation at the edge, reducing the energy and carbon footprint of AI workloads. While we acknowledge the possibility of unintended uses, we do not identify any specific societal risks that need to be highlighted in this context.

References

Agarwal, S., Plimpton, S. J., Hughart, D. R., Hsia, A. H., Richter, I., Cox, J. A., James, C. D., and Marinella, M. J. Resistive memory device requirements for a neural algorithm accelerator. In *International Joint Conference on*

- Neural Networks*, pp. 929–938. IEEE, 2016.
- Bottou, L., Curtis, F. E., and Nocedal, J. Optimization methods for large-scale machine learning. *SIAM review*, 60(2):223–311, 2018.
- Brown, T., Mann, B., Ryder, N., Subbiah, M., Kaplan, J. D., Dhariwal, P., Neelakantan, A., Shyam, P., Sastry, G., Askell, A., et al. Language models are few-shot learners. *Advances in neural information processing systems*, 33: 1877–1901, 2020.
- Burr, G. W., Shelby, R. M., Sidler, S., Di Nolfo, C., Jang, J., Boybat, I., Shenoy, R. S., Narayanan, P., Virwani, K., Giacometti, E. U., et al. Experimental demonstration and tolerancing of a large-scale neural network (165 000 synapses) using phase-change memory as the synaptic weight element. *IEEE Transactions on Electron Devices*, 62(11):3498–3507, 2015.
- Burr, G. W., BrightSky, M. J., Sebastian, A., Cheng, H.-Y., Wu, J.-Y., Kim, S., Sosa, N. E., Papandreou, N., Lung, H.-L., Pozidis, H., Eleftheriou, E., and Lam, C. H. Recent Progress in Phase-Change Memory Technology. *IEEE Journal on Emerging and Selected Topics in Circuits and Systems*, 6(2):146–162, 2016. ISSN 2156-3357.
- Chen, P., Lin, B., Wang, I.-T., Hou, T., Ye, J., Vruthula, S., Seo, J.-s., Cao, Y., and Yu, S. Mitigating effects of non-ideal synaptic device characteristics for on-chip learning. In *IEEE/ACM International Conference on Computer-Aided Design*, pp. 194–199. IEEE, 2015.
- Chen, T., Sun, Y., and Yin, W. Closing the gap: Tighter analysis of alternating stochastic gradient methods for bilevel problems. In *Proc. Advances in Neural Information Processing Systems*, virtual, 2021.
- Cosemans, S., Verhoef, B.-E., Dovenspeck, J., Papistas, I. A., Catthoor, F., Debacker, P., Mallik, A., and Verkest, D. Towards 10000TOPS/W DNN inference with analog in-memory computing – a circuit blueprint, device options and requirements. In *IEEE International Electron Devices Meeting*, pp. 22.2.1–22.2.4, 2019.
- Deaville, P., Zhang, B., Chen, L.-Y., and Verma, N. A maximally row-parallel MRAM in-memory-computing macro addressing readout circuit sensitivity and area. In *IEEE European Solid State Circuits Conference*, pp. 75–78. IEEE, 2021.
- Deng, J., Dong, W., Socher, R., Li, L.-J., Li, K., and Fei-Fei, L. Imagenet: A large-scale hierarchical image database. In *2009 IEEE Conference on Computer Vision and Pattern Recognition*, pp. 248–255. IEEE, 2009.
- Esmaeilzadeh, H., Sampson, A., Ceze, L., and Burger, D. Neural acceleration for general-purpose approximate programs. In *45th annual IEEE/ACM international symposium on microarchitecture*, pp. 449–460. IEEE, 2012.
- Franceschi, L., Donini, M., Frasconi, P., and Pontil, M. Forward and reverse gradient-based hyperparameter optimization. In *Proc. International Conference on Machine Learning*, Sydney, Australia, 2017.
- Franceschi, L., Frasconi, P., Salzo, S., Grazzi, R., and Pontil, M. Bilevel programming for hyperparameter optimization and meta-learning. In *Proc. International Conference on Machine Learning*, Stockholm, Sweden, 2018.
- Ghadimi, S. and Wang, M. Approximation methods for bilevel programming. *arXiv preprint arXiv:1802.02246*, 2018.
- Gokmen, T. Enabling training of neural networks on noisy hardware. *Frontiers in Artificial Intelligence*, 4:1–14, 2021.
- Gokmen, T. and Haensch, W. Algorithm for training neural networks on resistive device arrays. *Frontiers in Neuroscience*, 14, 2020.
- Gokmen, T. and Vlasov, Y. Acceleration of deep neural network training with resistive cross-point devices: Design considerations. *Frontiers in neuroscience*, 10:333, 2016.
- Gong, N., Rasch, M., Seo, S.-C., Gasasira, A., Solomon, P., Bragaglia, V., Consiglio, S., Higuchi, H., Park, C., Brew, K., and others. Deep learning acceleration in 14nm cmos compatible rram array: device, material and algorithm co-optimization. In *2022 International Electron Devices Meeting*, 2022a.
- Gong, N., Rasch, M., Seo, S.-C., Gasasira, A., Solomon, P., Bragaglia, V., Consiglio, S., Higuchi, H., Park, C., Brew, K., et al. Deep learning acceleration in 14nm CMOS compatible ReRAM array: device, material and algorithm co-optimization. In *IEEE International Electron Devices Meeting*, 2022b.
- He, K., Zhang, X., Ren, S., and Sun, J. Deep residual learning for image recognition. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 770–778, 2016.
- Hong, M., Wai, H.-T., Wang, Z., and Yang, Z. A Two-Timescale Framework for Bilevel Optimization: Complexity Analysis and Application to Actor-Critic. *arXiv preprint:2007.05170*, 2020.
- Huang, S., Sun, X., Peng, X., Jiang, H., and Yu, S. Overcoming challenges for achieving high in-situ training accuracy with emerging memories. In *Design, Automation &*

- Test in Europe Conference & Exhibition, pp. 1025–1030. IEEE, 2020.
- Jain, S. et al. Neural network accelerator design with resistive crossbars: Opportunities and challenges. *IBM Journal of Research and Development*, 63(6):10–1, 2019.
- Jang, J.-W., Park, S., Jeong, Y.-H., and Hwang, H. ReRAM-based synaptic device for neuromorphic computing. In *IEEE International Symposium on Circuits and Systems*, pp. 1054–1057, 2014.
- Jang, J.-W., Park, S., Burr, G. W., Hwang, H., and Jeong, Y.-H. Optimization of conductance change in $\text{Pr}_{1-x}\text{Ca}_x\text{MnO}_3$ -based synaptic devices for neuromorphic systems. *IEEE Electron Device Letters*, 36(5):457–459, 2015.
- Ji, K., Yang, J., and Liang, Y. Bilevel optimization: Convergence analysis and enhanced design. In *Proc. International Conference on Machine Learning*, virtual, 2021.
- Jouppi, N., Kurian, G., Li, S., Ma, P., Nagarajan, R., Nai, L., Patil, N., Subramanian, S., Swing, A., Towles, B., et al. TPU v4: An optically reconfigurable supercomputer for machine learning with hardware support for embeddings. In *Annual International Symposium on Computer Architecture*, pp. 1–14, 2023.
- Karimi, H., Nutini, J., and Schmidt, M. Linear convergence of gradient and proximal-gradient methods under the polyak-łojasiewicz condition. In *Joint European conference on machine learning and knowledge discovery in databases*, pp. 795–811. Springer, 2016.
- Kim, H., Rasch, M. J., Gokmen, T., Ando, T., Miyazoe, H., Kim, J.-J., Rozen, J., and Kim, S. Zero-shifting technique for deep neural network training on resistive cross-point arrays. *arXiv preprint arXiv:1907.10228*, 2019.
- Krizhevsky, A., Hinton, G., et al. Learning multiple layers of features from tiny images. 2009.
- Le Gallo, M. and Sebastian, A. An overview of phase-change memory device physics. *Journal of Physics D: Applied Physics*, 53(21):213002, 2020.
- LeCun, Y., Bottou, L., Bengio, Y., and Haffner, P. Gradient-based learning applied to document recognition. *Proceedings of the IEEE*, 86(11):2278–2324, 1998a.
- LeCun, Y., Corinna, C., and Burges, C. J. C. The MNIST database of handwritten digits. <http://yann.lecun.com/exdb/mnist>, 1998b.
- Lee, J.-H., Lim, D.-H., Jeong, H., Ma, H., and Shi, L. Exploring cycle-to-cycle and device-to-device variation tolerance in mlc storage-based neural network training. *IEEE Transactions on Electron Devices*, 66(5):2172–2178, 2019.
- Li, J., Wu, Z., Liu, G., Gokmen, T., and Chen, T. In-memory training on analog devices with limited conductance states via multi-tile residual learning. *arXiv preprint arXiv:2510.02516*, 2025.
- Maclaurin, D., Duvenaud, D., and Adams, R. Gradient-based hyperparameter optimization through reversible learning. In *Proc. International Conference on Machine Learning*, Lille, France, 2015.
- Modha, D. S., Akopyan, F., Andreopoulos, A., Appuswamy, R., Arthur, J. V., Cassidy, A. S., Datta, P., DeBole, M. V., Esser, S. K., Otero, C. O., et al. Neural inference at the frontier of energy, space, and time. *Science*, 382(6668): 329–335, 2023.
- Nandakumar, S. R., Le Gallo, M., Piveteau, C., Joshi, V., Mariani, G., Boybat, I., Karunaratne, G., Khaddam-Aljameh, R., Egger, U., Petropoulos, A., Antonakopoulos, T., Rajendran, B., Sebastian, A., and Eleftheriou, E. Mixed-precision deep learning based on computational memory. *Frontiers in Neuroscience*, 14, 2020.
- Onen, M., Emond, N., Wang, B., Zhang, D., Ross, F. M., Li, J., Yildiz, B., and Del Alamo, J. A. Nanosecond protonic programmable resistors for analog deep learning. *Science*, 377(6605):539–543, 2022.
- Papistas, I. A., Cosemans, S., Rooseleer, B., Doevenspeck, J., Na, M.-H., Mallik, A., Debacker, P., and Verkest, D. A 22 nm, 1540 TOP/s/W, 12.1 TOP/s/mm² in-memory analog matrix-vector-multiplier for DNN acceleration. In *IEEE Custom Integrated Circuits Conference*, pp. 1–2. IEEE, 2021.
- Pedregosa, F. Hyperparameter optimization with approximate gradient. In *Proc. International Conference on Machine Learning*, New York City, NY, 2016.
- Proakis, J. G. *Digital signal processing: principles, algorithms, and applications*, 4/E. Pearson Education India, 2007.
- Rao, M., Tang, H., Wu, J., Song, W., Zhang, M., Yin, W., Zhuo, Y., Kiani, F., Chen, B., Jiang, X., et al. Thousands of conductance levels in memristors integrated on CMOS. *Nature*, 615(7954):823–829, 2023.
- Rasch, M. J., Moreda, D., Gokmen, T., Le Gallo, M., Carta, F., Goldberg, C., El Maghraoui, K., Sebastian, A., and Narayanan, V. A flexible and fast pytorch toolkit for simulating training and inference on analog crossbar arrays. In *2021 IEEE 3rd international conference on artificial intelligence circuits and systems*, pp. 1–4. IEEE, 2021.

- Rasch, M. J., Carta, F., Fagbohungbe, O., and Gokmen, T. Fast offset corrected in-memory training. *arXiv preprint arXiv:2303.04721*, 2023.
- Rasch, M. J., Carta, F., Fagbohungbe, O., and Gokmen, T. Fast and robust analog in-memory deep neural network training. *Nature Communications*, 15(1):7133–7147, 2024.
- Sharma, D., Rath, S. P., Kundu, B., Korkmaz, A., Thompson, D., Bhat, N., Goswami, S., Williams, R. S., and Goswami, S. Linear symmetric self-selecting 14-bit kinetic molecular memristors. *Nature*, 633(8030):560–566, 2024.
- Shen, H. and Chen, T. A single-timescale analysis for stochastic approximation with multiple coupled sequences. In *Proc. Advances in Neural Information Processing Systems*, New Orleans, LA, 2022.
- Simonyan, K. and Zisserman, A. Very deep convolutional networks for large-scale image recognition. In *Proc. International Conference on Machine Learning*, 2014.
- Stecconi, T., Bragaglia, V., Rasch, M. J., Carta, F., Horst, F., Falcone, D. F., Ten Kate, S. C., Gong, N., Ando, T., Olziersky, A., et al. Analog resistive switching devices for training deep neural networks with the novel Tiki-Taka algorithm. *Nano Letters*, 24(3):866–872, 2024.
- Tang, J., Bishop, D., Kim, S., Copel, M., Gokmen, T., Todorov, T., Shin, S., Lee, K.-T., Solomon, P., Chan, K., et al. ECRAM as scalable synaptic cell for high-speed, low-power neuromorphic computing. In *IEEE International Electron Devices Meeting*, pp. 13–1. IEEE, 2018.
- Touvron, H., Lavril, T., Izacard, G., Martinet, X., Lachaux, M.-A., Lacroix, T., Rozière, B., Goyal, N., Hambro, E., Azhar, F., et al. Llama: Open and efficient foundation language models. *arXiv preprint arXiv:2302.13971*, 2023.
- Wan, W., Kubendran, R., Schaefer, C., Eryilmaz, S. B., Zhang, W., Wu, D., Deiss, S., Raina, P., Qian, H., Gao, B., et al. A compute-in-memory chip based on resistive random-access memory. *Nature*, 608(7923):504–512, 2022.
- Wang, Y., Wu, S., Tian, L., and Shi, L. SSM: a high-performance scheme for in situ training of imprecise memristor neural networks. *Neurocomputing*, 407:270–280, 2020.
- Wang, Z., Li, C., Lin, P., Rao, M., Nie, Y., Song, W., Qiu, Q., Li, Y., Yan, P., Strachan, J. P., et al. In situ training of feed-forward and recurrent convolutional memristor networks. *Nature Machine Intelligence*, 1(9):434–442, 2019.
- Wu, Z., Gokmen, T., Rasch, M. J., and Chen, T. Towards exact gradient-based training on analog in-memory computing. In *Advances in Neural Information Processing Systems*, 2024.
- Wu, Z., Xiao, Q., Gokmen, T., Fagbohungbe, O., and Chen, T. Analog in-memory training on general non-ideal resistive elements: The impact of response functions. In *Proc. Advances in Neural Information Processing Systems*, San Diego, CA, 2025.
- Yang, S., Wang, M., and Fang, E. X. Multilevel stochastic gradient methods for nested composition optimization. *SIAM Journal on Optimization*, 29(1):616–659, 2019.
- Yao, P., Wu, H., Gao, B., Eryilmaz, S. B., Huang, X., Zhang, W., Zhang, Q., Deng, N., Shi, L., Wong, H.-S. P., et al. Face classification using electronic synapses. *Nature communications*, 8(1):15199, 2017.
- Yao, P., Wu, H., Gao, B., Tang, J., Zhang, Q., Zhang, W., Yang, J. J., and Qian, H. Fully hardware-implemented memristor convolutional neural network. *Nature*, 577(7792):641–646, 2020.
- Zhang, B., Deaville, P., and Verma, N. Statistical computing framework and demonstration for in-memory computing systems. In *ACM/IEEE Design Automation Conference*, pp. 979–984, 2022.

Appendix for “Dynamic Symmetric Point Tracking: Tackling Non-ideal Reference in Analog In-memory Training”

Table of Contents

A	Notations and Preliminaries	13
B	Discussion of Algorithms	14
B.1	Two-stage analog training approach: Residual Learning with ZS algorithm	14
B.2	Comparison of E-RIDER, Residual Learning/TT-v2 and AGAD	14
B.3	Failure of dynamic zero-shifting algorithm design.	15
C	Proof of Convergence Rate of Algorithm 1 and Cyclic Version	15
C.1	Proof of Theorem 2.2: Convergence rate of Algorithm 1	15
C.2	Proof of Theorem C.2: Last-iterate convergence of Algorithm 1	15
C.3	Proof of Theorem C.3: Convergence rate of Algorithm 1, cyclic version	17
C.4	Proof of Theorem C.4: Last-iterate convergence of Algorithm 1, cyclic version	18
D	Proof of Auxiliary Lemmas	19
D.1	Proof of Lemma 3.5	19
D.2	Proof of Lemma 3.10	20
E	Proof of Theorem 3.7: Convergence of Algorithm 2	20
E.1	Main proof	20
E.2	Proof of Lemma E.1: Descent of sequence $\bar{W}_k$	25
E.3	Proof of Lemma E.2: Descent of sequence W_k	28
E.4	Proof of Lemma E.3: Descent of accumulated asymmetric sequence $\varphi(P_k)$	30
F	Additional Experiments and Experimental Setups	31
F.1	Device implementation details	31
F.2	Details of implementation for Figure 1 and Figure 2	32
F.3	Hyperparameter settings for Tables 1 and 2	33
F.4	Hyperparameter settings for Figure 4	34
F.5	Larger-scale ImageNet-1K fine-tuning experiment	34
F.6	Ablation studies of key hyperparameters	35

A. Notations and Preliminaries

In this section, we define a series of notations that will be used in the analysis.

Pseudo-inverse of diagonal matrix or vector. For a given diagonal matrix $U \in \mathbb{R}^{D \times D}$ with its d -th diagonal element $[U]_d$, we define the pseudo-inverse of a diagonal matrix U as $U^\dagger$, which is also a diagonal matrix with its d -th diagonal element

$$[U^\dagger]_d := \begin{cases} 1/[U]_d, & [U]_d \neq 0, \\ 0, & [U]_d = 0. \end{cases} \quad (19)$$

By definition, the pseudo-inverse satisfies $UU^\dagger V = U^\dagger UV$ for any diagonal matrix $U \in \mathbb{R}^D$ and any matrix $V \in \mathbb{R}^D$. With a slight abuse of notation, we also define the pseudo-inverse of a vector $W \in \mathbb{R}^D$ as $W^\dagger := \text{diag}(W)^\dagger$.

Algorithm 4 Two-stage analog training algorithm

- 1: **Inputs:** initialization P_0, Q_0, W_0 ; residual parameter γ ; response granularity $F_p(\cdot), F_w(\cdot), G_p(\cdot), G_w(\cdot)$
 - 2: estimate $\hat{W}^\diamond$ by Algorithm 1 with N number of pulses
 - 3: **for** $k = 0, 1, \dots, K - 1$ **do**
 - 4: evaluate $\bar{W}_k = W_k + \gamma(P_k - Q_k)$
 - 5: sample stochastic gradient $\nabla f(\bar{W}_k; \xi_k)$
 - 6: update P_{k+1} on analog device via (11a)
 - 7: set $Q_{k+1} = \hat{W}^\diamond$
 - 8: update W_{k+1} on analog device via (11b)
 - 9: **end for**
 - 10: **outputs:** $\{P_K, Q_K, W_K\}$
-

Weighted norm. For a weight $M \in \mathbb{R}_+^D$, the weighted norm $\|\cdot\|_M$ of $W \in \mathbb{R}^D$ is defined by

$$\|W\|_M := \sqrt{\sum_{d=1}^D [M]_d [W]_d^2} = \sqrt{W^\top \text{Diag}(M) W} \quad (20)$$

where $\text{Diag}(M) \in \mathbb{R}^{D \times D}$ rearranges the vector $M \in \mathbb{R}^D$ into a diagonal matrix.

Lemma A.1 ((Wu et al., 2025, Lemma 1)). $\|W\|_M$ has the following properties: (a) $\|W\|_M = \|W \odot \sqrt{M}\|$; (b) $\|W\|_M \leq \|W\| \sqrt{\|M\|_\infty}$; (c) $\|W\|_M \geq \|W\| \sqrt{\min\{[M]_d : k \in [K], d \in [D]\}}$.

Lemma A.2 ((Wu et al., 2025, Lemma 2)). Consider response functions in Definition 2.1, the increment defined in (2) is Lipschitz continuous with respect to ΔW under any weighted norm $\|\cdot\|_M$, i.e., for any $W, \Delta W, \Delta W' \in \mathbb{R}^D$ and $M \in \mathbb{R}_+^D$, it holds

$$\|\Delta W \odot F(W) - |\Delta W| \odot G(W) - (\Delta W' \odot F(W) - |\Delta W'| \odot G(W))\|_M \leq q_{\max} \|\Delta W - \Delta W'\|_M.$$

B. Discussion of Algorithms

In this section, we discuss some implementation details of algorithms.

B.1. Two-stage analog training approach: Residual Learning with ZS algorithm

The two-stage analog training algorithm consists of an independent SP estimation stage that uses Algorithm 1 to obtain a static SP estimate $\hat{W}^\diamond$, and an independent training stage that applies residual learning with $\hat{W}^\diamond$ fixed; i.e. we remove the Q_k -sequence update in Algorithm 2 and set $Q_k \equiv \hat{W}^\diamond$. We give a pseudo-code for this algorithm in Algorithm 4.

B.2. Comparison of E-RIDER, Residual Learning/TT-v2 and AGAD

The proposed E-RIDER has a similar form of AGAD (Rasch et al., 2023). However, AGAD uses the gradient $\nabla f(W_k; \xi_k)$ that are solely computed on the main array W_k . Instead, E-RIDER computes gradient on a mixed weight $\bar{W}_k = W_k + \gamma c_k(P_k - Q_k)$ so that achieves better performance in simulation. The key reason is that introducing the residual term $\gamma c_k(P_k - Q_k)$ with a nonzero γ effectively rescales the update, yielding a finer effective dynamic range and granularity than the raw device response. Moreover, this zero-shifting vector mitigates update asymmetry through the bilevel optimization. In simulation, we show that E-RIDER which uses the gradient evaluated at the mixed weight improves test accuracy, especially when the SP is substantially nonzero.

Compared with Residual Learning (Wu et al., 2025), E-RIDER uses a moving average sequence to track the SP and adds a chopper mechanism to amplify the SP tracking performance. From a theoretical perspective, Residual Learning fails to converge when the SP is nonzero, whereas E-RIDER with $p = 0$ is guaranteed to converge in our paper. Empirically, Residual Learning performs similarly to TT-v2, and both suffer substantial performance degradation under nonzero SP.

B.3. Failure of dynamic zero-shifting algorithm design.

From an optimization perspective, to balance the pulse overhead of SP estimation and model training, a natural approach is to integrate Algorithm 1 into analog training algorithms via multi-sequence updates (Yang et al., 2019; Shen & Chen, 2022), enabling dynamic SP estimation during the model training process. However, this approach is infeasible in analog training because each update sequence is executed on a different device, and the SP is device-specific. As a result, the SP tracked by the zero-shifting Algorithm 1 cannot be directly transferred to the sequence used for model training.

C. Proof of Convergence Rate of Algorithm 1 and Cyclic Version

C.1. Proof of Theorem 2.2: Convergence rate of Algorithm 1

Theorem 2.2 (Convergence rate of Algorithm 1). *Considering the response functions in Definition 2.1, then the iterates given by Algorithm 1 with N pulses satisfy*

$$\frac{1}{N} \sum_{n=0}^{N-1} \mathbb{E} [\|G(W_n)\|^2] \leq \mathcal{O} \left(\frac{1}{N \Delta w_{\min}} \right) + \Theta(\Delta w_{\min}).$$

Theorem 2.2. Define the following notations:

$$\varphi(W) := \int_{W^\diamond}^W G(W') dW', \quad \nabla \varphi(W) := G(W), \quad L_q := \max_W \Delta G(W). \quad (21)$$

Under gradient boundedness implied by Definition 2.1, it holds that:

$$\begin{aligned} \varphi(W_{n+1}) &\leq \varphi(W_n) + \langle \nabla \varphi(W_n), W_{n+1} - W_n \rangle + \frac{L_q}{2} \|W_{n+1} - W_n\|^2 \\ &= \varphi(W_n) + \langle G(W_n), \varepsilon_n F(W_n) - |\varepsilon_n| G(W_n) \rangle + \frac{L_q}{2} \|\varepsilon_n F(W_n) - |\varepsilon_n| G(W_n)\|^2. \end{aligned} \quad (22)$$

where L_q is the gradient boundedness constant implied by Definition 2.1. The equality holds by substituting the stochastic updating equation (7). Taking expectation over ε_n on both sides, we get:

$$\begin{aligned} \mathbb{E}[\varphi(W_{n+1}) \mid W_n] &\leq \varphi(W_n) - \mathbb{E}[|\varepsilon_n|] \|G(W_n)\|^2 + \frac{L_q}{2} \Delta w_{\min}^2 q_{\max}^2 \\ &= \varphi(W_n) - \Delta w_{\min} \|G(W_n)\|^2 + \frac{L_q}{2} \Delta w_{\min}^2 q_{\max}^2 \\ &\leq \varphi(W_n) - \Delta w_{\min} \|G(W_n)\|^2 + \frac{L_q}{2} \Delta w_{\min}^2 q_{\max}^2. \end{aligned} \quad (23)$$

Taking total expectation $\mathcal{F}_n$ to both sides of (23) and using $\mathbb{E}[\mathbb{E}[\varphi(W_{n+1}) \mid \mathcal{F}_n]] = \mathbb{E}[\varphi(W_{n+1})]$, we get:

$$\mathbb{E}[\varphi(W_{n+1})] \leq \mathbb{E}[\varphi(W_n)] - \Delta w_{\min} \mathbb{E}[\|G(W_n)\|^2] + \frac{L_q}{2} \Delta w_{\min}^2 q_{\max}^2. \quad (24)$$

Taking average over all n from 0 to $N-1$, we get:

$$\frac{1}{N} \sum_{n=0}^{N-1} \mathbb{E} [\|G(W_n)\|^2] \leq \frac{\varphi(W_0) - \varphi(W^*)}{N \Delta w_{\min}} + \frac{L_q}{2} \Delta w_{\min} q_{\max}^2 \quad (25)$$

which completes the proof. $\square$

C.2. Proof of Theorem C.2: Last-iterate convergence of Algorithm 1

To get the enhanced last-iterate pulse complexity for Algorithm 1, we focus on the family of devices with the following monotone response functions.

Definition C.1 (Monotone response functions). Response functions $q_+(\cdot)$ and $q_-(\cdot)$ are said to be monotone with modulus $\mu_q > 0$ if

$$\nabla q_-(w) \geq \mu_q, \quad \text{and} \quad \nabla q_+(w) \leq -\mu_q. \quad (26)$$

Definition C.1 suggests that $q_-(\cdot)$ is monotonically increasing and $q_+(\cdot)$ is monotonically decreasing. The response functions of a wide range of AIMC devices, such as linear, exponential, and power device, satisfy Definition C.1 (Wu et al., 2025). With Definition C.1, $G(W)$ is strongly monotone with μ_q so that we have the following last-iterates convergence.

Theorem C.2 (Last-iterate convergence of Algorithm 1). *Considering the response functions satisfying Definition 2.1 and C.1 and assuming $\mu_g < \frac{1}{2\Delta w_{\min}}$, then to achieve $\mathbb{E}[\|W^N - W^\diamond\|^2] \leq \delta$, we need*

$$N \leq \frac{1}{2\mu_q \Delta w_{\min}} \log \left(\frac{2\|W^0 - W^\diamond\|^2}{\delta} \right).$$

and the minimal achievable error δ is $\delta = \frac{2q_{\max}^2 \Delta w_{\min}}{\mu_q}$.

From Theorem C.2, we know that the zero-shifting Algorithm 1 achieves a sublinear dependence on the target error δ , and that the required number of pulses scales linearly with $1/\Delta w_{\min}$. In practice, the response granularity $\Delta w_{\min}$ of AIMC device is made sufficiently small to ensure the gradient conversion precision (Rao et al., 2023; Sharma et al., 2024), so that the computational complexity of Algorithm 1 is relatively high.

Theorem C.2. We begin the proof by one step descent form stochastic updating equation (7) and taking expectation over ε_n on both sides:

$$\begin{aligned} & \mathbb{E}[\|W_{n+1} - W^\diamond\|^2 \mid W_n] \\ &= \mathbb{E}[\|W_n - \Delta w_{\min} \varepsilon_n F(W_n) - \Delta w_{\min} G(W_n) - W^\diamond\|^2 \mid W_n] \\ &= \|W_n - W^\diamond\|^2 - 2\Delta w_{\min} \langle W_n - W^\diamond, G(W_n) - G_w(W^\diamond) \rangle + \Delta w_{\min}^2 \mathbb{E}[\|\varepsilon_n F(W_n) + G(W_n)\|^2 \mid W_n] \\ &\leq (1 - 2\Delta w_{\min} \mu_q) \|W_n - W^\diamond\|^2 + 2\Delta w_{\min}^2 q_{\max}^2. \end{aligned} \quad (27)$$

The second equality holds for $\mathbb{E}[\varepsilon_n F(W_n)] = 0$ and $G_w(W^\diamond) = 0$. The inequality holds for the strongly convex assumptions in definition C.1 and bounded response functions in definition 2.1:

$$\langle W_n - W^\diamond, G(W_n) - G_w(W^\diamond) \rangle \geq \mu_q \|W_n - W^\diamond\|^2, \quad \mathbb{E}[\|\varepsilon_n F(W_n) + G(W_n)\|^2 \mid W_n] \leq 2q_{\max}^2.$$

Taking total expectation $\mathcal{F}_n$ to both sides of (27) and iterating this recursion for $n = 0, \dots, N-1$ gives:

$$\begin{aligned} \mathbb{E}[\|W_N - W^\diamond\|^2] &\leq (1 - 2\Delta w_{\min} \mu_q)^N \|W_0 - W^\diamond\|^2 + 2\Delta w_{\min}^2 q_{\max}^2 \sum_{n=0}^{N-1} (1 - 2\Delta w_{\min} \mu_q)^n \\ &\leq (1 - 2\Delta w_{\min} \mu_q)^N \|W_0 - W^\diamond\|^2 + \frac{\Delta w_{\min} q_{\max}^2}{\mu_q}. \end{aligned} \quad (28)$$

To ensure that the error bound is below a given tolerance δ , we split the upper bound into two terms and require each of them to be at most $\frac{\delta}{2}$, which means $\frac{\Delta w_{\min} q_{\max}^2}{\mu_q} \leq \frac{\delta}{2}$ and $(1 - 2\mu_q \Delta w_{\min})^N \|W_0 - W^\diamond\|^2 \leq \frac{\delta}{2}$.

The first inequality holds when $\Delta w_{\min} \leq \frac{\delta \mu_q}{2q_{\max}^2}$. Taking logarithms on both sides of the second inequality gives:

$$\log \left(\frac{2\|W_0 - W^\diamond\|^2}{\delta} \right) \leq N \log \left(\frac{1}{1 - 2\mu_q \Delta w_{\min}} \right). \quad (29)$$

Using the inequality $\log(\frac{1}{p}) \geq 1 - p$, when $\mu_q < \frac{1}{2\Delta w_{\min}}$, it suffices to require

$$N \geq \frac{1}{2\mu_q \Delta w_{\min}} \log \left(\frac{2\|W_0 - W^\diamond\|^2}{\delta} \right) \quad (30)$$

which completes the proof. $\square$

C.3. Proof of Theorem C.3: Convergence rate of Algorithm 1, cyclic version

In this section, we model the empirical implementation of the zero-shifting technique, where cyclic alternating pulses are applied. We also establish a convergence guarantee analogous to Theorem 2.2. We first show the update dynamics below:

$$\begin{aligned} W_{2n} &= W_{2n-1} - \Delta w_{\min} F(W_{2n-1}) - \Delta w_{\min} G(W_{2n-1}), \\ W_{2n+1} &= W_{2n} + \Delta w_{\min} F(W_{2n}) - \Delta w_{\min} G(W_{2n}). \end{aligned} \quad (31)$$

Theorem C.3 (Convergence rate of Algorithm 1, cyclic version). *Considering the response functions satisfying Definition 2.1, then when $N = \mathcal{O}(\Delta w_{\min}^{-2} \kappa_2^{-2})$, it holds that*

$$\frac{1}{2N} \sum_{n=0}^{2N-1} \|G(W_n)\|^2 \leq \frac{\varphi(W_0) - \varphi^*}{2N \Delta w_{\min}} + \frac{3 \Delta w_{\min} q_{\max}^2 L_q}{2} \quad (32)$$

where L_q is the gradient boundedness constant implied by Definition 2.1. Therefore, the minimal achievable SP estimation error $\frac{1}{2N} \sum_{n=0}^{2N-1} \mathbb{E} [\|G(W_n)\|^2] \leq \delta$ is $\delta = \mathcal{O}(\Delta w_{\min})$. To achieve error $\delta \geq \mathcal{O}(\Delta w_{\min})$, it requires $N = \mathcal{O}(\frac{1}{\delta \Delta w_{\min}})$ number of pulses.

Theorem C.3. We begin the proof by using the same notations defined in equation (21). Under gradient boundedness implied by Definition 2.1, it holds that:

$$\begin{aligned} \varphi(W_{n+1}) &\leq \varphi(W_n) + \langle \nabla \varphi(W_n), W_{n+1} - W_n \rangle + \frac{L_q}{2} \|W_{n+1} - W_n\|^2 \\ &= \varphi(W_n) + \Delta w_{\min} \langle G(W_n), F(W_n) - G(W_n) \rangle + \frac{L_q}{2} \Delta w_{\min}^2 \|F(W_n) - G(W_n)\|^2 \\ &\leq \varphi(W_n) - \Delta w_{\min} \|G(W_n)\|^2 + \Delta w_{\min} \langle G(W_n), F(W_n) \rangle + \frac{L_q}{2} \Delta w_{\min}^2 q_{\max}^2. \end{aligned} \quad (33)$$

The second equality holds by substituting the stochastic updating equation (31). The inequality holds for $F(W_n) - G(W_n) = \frac{q_+(W_n) + q_-(W_n)}{2} - \frac{q_+(W_n) - q_-(W_n)}{2} = q_+(W_n) \leq q_{\max}$. Similarly we have:

$$\begin{aligned} \varphi(W_{n+2}) &\leq \varphi(W_{n+1}) + \Delta w_{\min} \langle G(W_{n+1}), -F(W_{n+1}) - G(W_{n+1}) \rangle + \frac{L_q}{2} \Delta w_{\min}^2 q_{\max}^2 \\ &= \varphi(W_{n+1}) - \Delta w_{\min} \|G(W_{n+1})\|^2 - \Delta w_{\min} \langle G(W_{n+1}), F(W_{n+1}) \rangle + \frac{L_q}{2} \Delta w_{\min}^2 q_{\max}^2. \end{aligned} \quad (34)$$

Combining (33) and (34), we get:

$$\begin{aligned} \varphi(W_{n+2}) &\leq \varphi(W_n) - \Delta w_{\min} \|G(W_n)\|^2 - \Delta w_{\min} \|G(W_{n+1})\|^2 \\ &\quad + \Delta w_{\min} \left(\langle G(W_n), F(W_n) \rangle - \langle G(W_{n+1}), F(W_{n+1}) \rangle \right) + L_q \Delta w_{\min}^2 q_{\max}^2 \\ &\leq \varphi(W_n) - \Delta w_{\min} \|G(W_n)\|^2 - \Delta w_{\min} \|G(W_{n+1})\|^2 + 3 \Delta w_{\min}^2 q_{\max}^2 L_q. \end{aligned} \quad (35)$$

The second inequality holds for

$$\begin{aligned} &\langle G(W_n), F(W_n) \rangle - \langle G(W_{n+1}), F(W_{n+1}) \rangle \\ &= \langle G(W_n), F(W_n) - F(W_{n+1}) \rangle - \langle G(W_{n+1}) - G(W_n), F(W_{n+1}) \rangle \\ &\leq \|G(W_n)\| \|F(W_n) - F(W_{n+1})\| + \|G(W_{n+1}) - G(W_n)\| \|F(W_{n+1})\| \\ &\leq 2q_{\max} L_q \|W_{n+1} - W_n\| \\ &\leq 2q_{\max} L_q \Delta w_{\min} q_{\max}. \end{aligned} \quad (36)$$

By summing the recursion in Eq. (35) over the iterations, we obtain:

$$\varphi(W_{2N}) \leq \varphi(W_0) - \sum_{n=0}^{2N-1} \Delta w_{\min} \|G(W_n)\|^2 + 3 \Delta w_{\min}^2 q_{\max}^2 L_q N. \quad (37)$$

Taking average, we get:

$$\frac{1}{2N} \sum_{n=0}^{2N-1} \|G(W_n)\|^2 \leq \frac{\varphi(W_0) - \varphi^*}{2N \Delta w_{\min}} + \frac{3 \Delta w_{\min} q_{\max}^2 L_q}{2}. \quad (38)$$

□

C.4. Proof of Theorem C.4: Last-iterate convergence of Algorithm 1, cyclic version

Theorem C.4 (Last-iterate convergence of Algorithm 1, cyclic version). *Considering the response functions satisfying Definition 2.1 and C.1 and assuming $\mu_q < \frac{1}{3\mu_q \Delta w_{\min}}$, then it holds that*

$$(1 - 2\Delta w_{\min} \mu_q)^{2N} \|W_0 - W^\diamond\|^2 + \frac{\Delta w_{\min} q_{\max}^2 \left(4 + \frac{L_q^2}{\mu_q} + 2\mu_q\right)}{\mu_q}.$$

Moreover, the minimal achievable error $\|W^{2N} - W^\diamond\|^2 \leq \delta$ is $\delta = \mathcal{O}(\Delta w_{\min})$, and to achieve $\|W^{2N} - W^\diamond\|^2 \leq \delta$ with $\delta \geq \mathcal{O}(\Delta w_{\min})$, one need the number of pulses satisfying

$$N \geq \frac{1}{4\mu_q \Delta w_{\min}} \log \left(\frac{2\|W^0 - W^\diamond\|^2}{\delta} \right).$$

Theorem C.4. We begin the proof by one step descent form cyclic updating equation (31)

$$\begin{aligned} \|W_{n+1} - W^\diamond\|^2 &= \|W_n - \Delta w_{\min} F(W_n) - \Delta w_{\min} G(W_n) - W^\diamond\|^2 \\ &= \|W_n - W^\diamond\|^2 - 2\Delta w_{\min} \langle W_n - W^\diamond, G(W_n) - G(W^\diamond) \rangle + \Delta w_{\min}^2 \|F(W_n) + G(W_n)\|^2 \\ &\quad - 2\Delta w_{\min} \langle W_n - W^\diamond, F(W_n) \rangle \\ &\leq (1 - 2\Delta w_{\min} \mu_q) \|W_n - W^\diamond\|^2 + 2\Delta w_{\min}^2 q_{\max}^2 - 2\Delta w_{\min} \langle W_n - W^\diamond, F(W_n) \rangle. \end{aligned} \quad (39)$$

The second equality holds for $G_w(W^\diamond) = 0$. The inequality holds for the strongly convex assumptions in definition C.1 and bounded response functions in definition 2.1:

$$\langle W_n - W^\diamond, G(W_n) - G_w(W^\diamond) \rangle \geq \mu_q \|W_n - W^\diamond\|^2, \quad \|F(W_n) + G(W_n)\|^2 \leq 2q_{\max}^2.$$

Similarly we have:

$$\begin{aligned} \|W_{n+2} - W^\diamond\|^2 &= \|W_{n+1} + \Delta w_{\min} F(W_{n+1}) - \Delta w_{\min} G(W_{n+1}) - W^\diamond\|^2 \\ &\leq (1 - 2\Delta w_{\min} \mu_q) \|W_{n+1} - W^\diamond\|^2 + 2\Delta w_{\min}^2 q_{\max}^2 + 2\Delta w_{\min} \langle W_{n+1} - W^\diamond, F(W_{n+1}) \rangle. \end{aligned} \quad (40)$$

The last term in (39) and (40) can be bounded by:

$$\begin{aligned} &2\Delta w_{\min} \langle W_n - W^\diamond, F(W_{n+1}) \rangle - 2\Delta w_{\min} \langle W_n - W^\diamond, F(W_n) \rangle (1 - 2\Delta w_{\min} \mu_q) \\ &\leq 2\Delta w_{\min} \langle W_n - W^\diamond, F(W_{n+1}) - F(W_n) \rangle + 4\Delta w_{\min}^2 \mu_q \|W_n - W^\diamond\| \|F(W_n)\| \\ &\leq \Delta w_{\min}^2 \mu_q \|W_n - W^\diamond\|^2 + \frac{\|F(W_{n+1}) - F(W_n)\|^2}{\mu_q} + 4\Delta w_{\min}^2 \mu_q \|W_n - W^\diamond\| q_{\max} \\ &\leq \Delta w_{\min}^2 \mu_q \|W_n - W^\diamond\|^2 + \frac{L_q^2 \Delta w_{\min}^2 q_{\max}^2}{\mu_q} + 2\Delta w_{\min}^2 \mu_q \|W_n - W^\diamond\|^2 + 2\Delta w_{\min}^2 \mu_q q_{\max}^2 \\ &\leq 3\Delta w_{\min}^2 \mu_q \|W_n - W^\diamond\|^2 + \frac{L_q^2 \Delta w_{\min}^2 q_{\max}^2}{\mu_q} + 2\Delta w_{\min}^2 \mu_q q_{\max}^2 \\ &\leq \Delta w_{\min} \mu_q \|W_n - W^\diamond\|^2 + \frac{L_q^2 \Delta w_{\min}^2 q_{\max}^2}{\mu_q} + 2\Delta w_{\min}^2 \mu_q q_{\max}^2. \end{aligned} \quad (41)$$

The first inequality follows by expanding the term and applying Cauchy–Schwarz. The second and fourth inequalities use Young’s inequality together with the bound $|F(W_n)| \leq q_{\max}$. The last step holds under the step-size condition $3\Delta w_{\min}^2 \mu_q \leq \Delta w_{\min} \mu_q$. Combining (39) and (40) and substituting (41), we get:

$$\begin{aligned} \|W_{n+2} - W^\diamond\|^2 &\leq (1 - 2\Delta w_{\min} \mu_q)^2 \|W_n - W^\diamond\|^2 + 4\Delta w_{\min}^2 q_{\max}^2 + \frac{L_q^2 \Delta w_{\min}^2 q_{\max}^2}{\mu_q} \\ &\quad + \Delta w_{\min} \mu_q \|W_n - W^\diamond\|^2 + 2\Delta w_{\min}^2 \mu_q q_{\max}^2 \\ &\leq (1 - \Delta w_{\min} \mu_q)^2 \|W_n - W^\diamond\|^2 + \Delta w_{\min}^2 q_{\max}^2 \left(4 + \frac{L_q^2}{\mu_q} + 2\mu_q\right). \end{aligned} \quad (42)$$

Iterating the recursion for $n = 0, \dots, 2N - 1$ gives:

$$\begin{aligned} \|W_{2N} - W^\diamond\|^2 &\leq (1 - 2\Delta w_{\min} \mu_q)^{2N} \|W_0 - W^\diamond\|^2 + \sum_{n=0}^{2N-1} (1 - 2\Delta w_{\min} \mu_q)^n \Delta w_{\min}^2 q_{\max}^2 \left(4 + \frac{L_q^2}{\mu_q} + 2\mu_q\right) \\ &\leq (1 - 2\Delta w_{\min} \mu_q)^{2N} \|W_0 - W^\diamond\|^2 + \frac{\Delta w_{\min} q_{\max}^2 \left(4 + \frac{L_q^2}{\mu_q} + 2\mu_q\right)}{\mu_q}. \end{aligned} \quad (43)$$

To ensure the total error remains within the tolerance δ , we bound each term in the upper bound by $\delta/2$. This yields the following two sufficient conditions:

$$\frac{\Delta w_{\min} q_{\max}^2 \left(4 + \frac{L_q^2}{\mu_q} + 2\mu_q\right)}{\mu_q} \leq \frac{\delta}{2} \quad \text{and} \quad (1 - 2\mu_q \Delta w_{\min})^{2N} \|W_0 - W^\diamond\|^2 \leq \frac{\delta}{2}. \quad (44)$$

The first condition implies a lower bound on the achievable tolerance, requiring $\delta \geq \frac{2q_{\max}^2(4+L_q^2/\mu_q+2\mu_q)\Delta w_{\min}}{\mu_q}$. For the second condition, rearranging terms and taking the logarithm yields

$$\log \left(\frac{2\|W_0 - W^\diamond\|^2}{\delta} \right) \leq 2N \log \left(\frac{1}{1 - 2\mu_q \Delta w_{\min}} \right). \quad (45)$$

Applying the inequality $\log(1/x) \geq 1 - x$ (valid for $x \in (0, 1]$) with $x = 1 - 2\mu_q \Delta w_{\min}$, we observe that the right-hand side is lower-bounded by $2N(2\mu_q \Delta w_{\min}) = 4N\mu_q \Delta w_{\min}$. Therefore, to satisfy the inequality, it suffices to set

$$N \geq \frac{1}{4\mu_q \Delta w_{\min}} \log \left(\frac{2\|W_0 - W^\diamond\|^2}{\delta} \right), \quad (46)$$

which completes the proof. $\square$

D. Proof of Auxiliary Lemmas

D.1. Proof of Lemma 3.5

Proof. According to the update of Q_k in (12), we know

$$Q_{k+1} - W^\diamond = (1 - \eta)(Q_k - W^\diamond) + \eta(P_{k+1} - W^\diamond).$$

Taking square norm of both sides yield

$$\begin{aligned} \|Q_{k+1} - W^\diamond\|^2 &= (1 - \eta)^2 \|Q_k - W^\diamond\|^2 + \eta^2 \|P_{k+1} - W^\diamond\|^2 + 2\eta(1 - \eta) \langle P_{k+1} - W^\diamond, Q_k - W^\diamond \rangle \\ &= (1 - \eta)^2 \|Q_k - W^\diamond\|^2 + \eta^2 \|P_{k+1} - W^\diamond\|^2 \\ &\quad + \eta(1 - \eta) \|P_{k+1} - W^\diamond\|^2 + \eta(1 - \eta) \|Q_k - W^\diamond\|^2 - \eta(1 - \eta) \|P_{k+1} - Q_k\|^2 \\ &= (1 - \eta) \|Q_k - W^\diamond\|^2 + \eta \|P_{k+1} - W^\diamond\|^2 - \eta(1 - \eta) \|P_{k+1} - Q_k\|^2 \end{aligned} \quad (47)$$

where the second equation is because $\langle A, B \rangle = \frac{1}{2}\|A\|^2 + \frac{1}{2}\|B\|^2 - \frac{1}{2}\|A - B\|^2$. On the other hand, letting $\cos(P_{k+1} - W^\diamond, P_{k+1} - Q_k) := \cos \theta$, we have

$$\|Q_k - W^\diamond\|^2 = \|P_{k+1} - W^\diamond\|^2 + \|P_{k+1} - Q_k\|^2 - 2\|P_{k+1} - W^\diamond\|\|P_{k+1} - Q_k\| \cos \theta. \quad (48)$$

Plugging (48) into (47), we get

$$\|Q_{k+1} - W^\diamond\|^2 = \|P_{k+1} - W^\diamond\|^2 + (1 - \eta)^2\|P_{k+1} - Q_k\|^2 - 2(1 - \eta)\|P_{k+1} - W^\diamond\|\|P_{k+1} - Q_k\| \cos \theta$$

Since $\cos \theta > 0$ implies $\|P_{k+1} - Q_k\| \neq 0$, then choosing $1 > \eta > \max \left\{ 1 - \frac{2\|P_{k+1} - W^\diamond\| \cos \theta}{\|P_{k+1} - Q_k\|}, 0 \right\}$ yields

$$\|Q_{k+1} - W^\diamond\|^2 < \|P_{k+1} - W^\diamond\|^2.$$

□

D.2. Proof of Lemma 3.10

Proof. From the digital signal processing perspective, the moving average update (12) defines a stable first-order infinite impulse response (IIS) low-pass filter from P_k to Q_k . Taking the z -transform of (12) yields

$$Q(z) = (1 - \eta)z^{-1}Q(z) + \eta P(z),$$

which gives the filter's transfer function

$$H(z) \triangleq \frac{Q(z)}{P(z)} = \frac{\eta}{1 - (1 - \eta)z^{-1}}. \quad (49)$$

The transfer function has a single pole at $z = 1 - \eta$ with the following magnitude of the frequency response

$$|H(e^{j\omega})|^2 = \frac{\eta^2}{1 + (1 - \eta)^2 - 2(1 - \eta) \cos \omega}. \quad (50)$$

□

E. Proof of Theorem 3.7: Convergence of Algorithm 2

This section provides the convergence analysis details of the Algorithm 2 under the strongly convexity condition.

Theorem 3.7 (Convergence of Algorithm 2). *Suppose Assumptions 3.1-3.4, 3.6 hold and the response functions F_p, G_p, F_w, G_w satisfy Definition 2.1. If $C_\star \geq \frac{4\sqrt{2}\sigma}{\mu} \left(\frac{q_{\max}}{q_{\min}} \right)^{\frac{3}{2}}$, let $\alpha = \Theta\left(\frac{1}{\sqrt{K}}\right)$, $\beta = \Theta(\alpha\gamma\mu)$, $\eta = \Theta(\alpha\mu)$, $\gamma = \Theta(1)$, it holds that*

$$E_K \leq \mathcal{O}\left(\frac{\kappa_1 \kappa_2^5}{\sqrt{K}}\right) + \Theta(\Delta w_{\min}). \quad (15)$$

where $\kappa_1 := L/\mu$ and $\kappa_2 := q_{\max}/q_{\min}$ are the condition number of the function and device.

E.1. Main proof

Theorem 3.7. Define the following notations $M_w(W_k) := Q_+(W_k) \odot Q_-(W_k) \in \mathbb{R}^D$, $M_p(P_k) := Q_+(P_k) \odot Q_-(P_k) \in \mathbb{R}^D$. The proof of the RIDER convergence relies on the following two lemmas, which provide the sufficient descent of W_k and $\bar{W}_k$, respectively.

Lemma E.1 (Descent lemma of lower-level problem). *Under Assumptions 3.1-3.2, it holds that*

$$\mathbb{E}_{\xi_k, b_k} [f(\bar{W}_{k+1})] \leq f(\bar{W}_k) - \frac{\alpha\gamma}{8q_{\max}} \|\nabla f(\bar{W}_k)\|_{M_p(P_k)}^2 + \frac{\alpha\gamma\sigma^2}{2} \left\| \frac{G_p(P_k)}{\sqrt{F_p(P_k)}} \right\|_\infty^2 + \Theta\left(\frac{\gamma\Delta w_{\min}}{q_{\min}}\right)$$

$$\begin{aligned}
 & + \frac{2q_{\max}\gamma}{\alpha} \mathbb{E}_{\xi_k} [\|Q_{k+1} - Q_k\|_{M_p(P_k)}^2] + \frac{3\gamma^2 L}{2} \mathbb{E}_{\xi_k} [\|Q_{k+1} - Q_k\|^2] \\
 & - \left(\frac{\gamma}{2\alpha q_{\max}} - 3\gamma^2 L \right) \mathbb{E}_{\xi_k, b_k} [\|P_{k+1} - P_k + \alpha(\nabla f(\bar{W}_k; \xi_k) - \nabla f(\bar{W}_k)) \odot F_p(P_k) - b_k\|^2] \\
 & + 3\alpha^2 \gamma^2 L q_{\max}^2 \sigma^2 + \frac{q_{\max}}{\alpha \gamma} \mathbb{E}_{\xi_k} [\|W_{k+1} - W_k\|_{M_p(P_k)}^2] + \frac{3L}{2} \mathbb{E}_{\xi_k} [\|W_{k+1} - W_k\|^2].
 \end{aligned} \tag{51}$$

Keeping W_k fixed, we also have

$$\begin{aligned}
 & \mathbb{E}_{\xi_k, b_k} [f(W_k + \gamma P_{k+1} - \gamma Q_k)] \\
 & \leq f(\bar{W}_k) - \frac{\alpha \gamma}{2q_{\max}} \|\nabla f(\bar{W}_k)\|_{M_p(P_k)}^2 + \frac{\alpha \gamma \sigma^2}{2} \left\| \frac{G_p(P_k)}{\sqrt{F_p(P_k)}} \right\|_{\infty}^2 + \Theta \left(\frac{\gamma \Delta w_{\min}}{q_{\min}} \right) \\
 & \quad - \left(\frac{\gamma}{2\alpha q_{\max}} - \gamma^2 L \right) \mathbb{E}_{\xi_k, b_k} [\|P_{k+1} - P_k + \alpha(\nabla f(\bar{W}_k; \xi_k) - \nabla f(\bar{W}_k)) \odot F_p(P_k) - b_k\|^2] + \alpha^2 \gamma^2 L q_{\max}^2 \sigma^2. \\
 & \leq f(\bar{W}_k) - \frac{\alpha \gamma}{2q_{\max}} \|\nabla f(\bar{W}_k)\|_{M_p(P_k)}^2 + \frac{\alpha \gamma \sigma^2}{2} \left\| \frac{G_p(P_k)}{\sqrt{F_p(P_k)}} \right\|_{\infty}^2 + \alpha^2 \gamma^2 L q_{\max}^2 \sigma^2 + \Theta \left(\frac{\gamma \Delta w_{\min}}{q_{\min}} \right)
 \end{aligned} \tag{52}$$

Recall that we use the square norm of $P^*(W, Q) - Q = (W^* - W)/\gamma$ to measure the convergence of the upper-level problem.

Lemma E.2 (Descent lemma of upper-level problem). *Under Assumption 2.1, it holds that*

$$\begin{aligned}
 \mathbb{E}_{b'_k} [\|P^*(W_{k+1}, Q_{k+1}) - Q_{k+1}\|^2] & \leq \|P^*(W_k, Q_k) - Q_k\|^2 - \frac{\beta}{2\gamma q_{\max}} \|P^*(W_k, Q_k) - Q_k\|_{M_w(W_k)}^2 + \Theta \left(\frac{\Delta w_{\min}}{\gamma q_{\max}} \right) \\
 & \quad - \frac{\beta}{2\gamma q_{\max}} \|(P^*(W_k, Q_k) - Q_k) \odot F_w(W_k) - |P^*(W_k, Q_k) - Q_k| \odot G_w(W_k)\|^2 \\
 & \quad + \frac{2\beta q_{\max}^3}{\gamma} \|P_{k+1} - P^*(W_k, Q_k)\|_{M_w(W_k)}^2 + \frac{2\beta^2}{\gamma^2} \|P_{k+1} - P^*(W_k, Q_k)\|^2.
 \end{aligned} \tag{53}$$

Lemma E.3 (Descent lemma of accumulated asymmetric sequence). *Under Assumption 2.1 and 3.6, it holds that*

$$\begin{aligned}
 \mathbb{E}_{\xi_k, b_k} [\varphi(P_{k+1})] & \leq \varphi(P_k) - \frac{\alpha C_{\star}}{2} \|G_p(P_k)\|^2 + \alpha^2 L q_{\max}^2 \sigma^2 + \frac{\alpha q_{\max}^2}{2C_{\star}} \|\nabla f(\bar{W}_k)\|^2 + \Theta \left(\frac{\Delta w_{\min}}{\gamma q_{\max}} \right) \\
 & \quad + L \mathbb{E}_{\xi_k, b_k} [\|P_{k+1} - P_k + \alpha(\nabla f(\bar{W}_k; \xi_k) - \nabla f(\bar{W}_k)) \odot F_p(P_k) - b_k\|^2].
 \end{aligned} \tag{54}$$

The proof of Lemma E.1, E.2 and E.3 are deferred to Appendices E.2, E.3 and E.4 respectively. For the response functions that satisfy Definition 2.1, we have

$$\min\{[M_w(W_k)]_d : d \in [D]\} \geq q_{\min}^2 > 0, \quad \text{and} \quad \min\{[M_p(P_k)]_d : d \in [D]\} \geq q_{\min}^2 > 0. \tag{55}$$

Now we deal with the weighted norm in two lemmas by Lemma A.1

$$\frac{\beta}{2\gamma q_{\max}} \|P^*(W_k, Q_k)\|_{M_p(P_k)}^2 \geq \frac{\beta q_{\min}^2}{2\gamma q_{\max}} \|P^*(W_k, Q_k)\|^2 \tag{56}$$

$$\frac{\alpha}{4q_{\max}} \|\nabla f(\bar{W}_k)\|_{M_p(P_k)}^2 \geq \frac{\alpha q_{\min}^2}{4q_{\max}} \|\nabla f(\bar{W}_k)\|^2 \tag{57}$$

$$\frac{q_{\max}}{\alpha} \|W_{k+1} - W_k\|_{M_p(P_k)}^2 \leq \frac{q_{\max}}{\alpha q_{\min}^2} \|W_{k+1} - W_k\|^2 \tag{58}$$

$$\frac{2\beta q_{\max}^3}{\gamma} \|P_{k+1} - P^*(W_k, Q_k)\|_{M_w(W_k)}^2 \leq \frac{2\beta q_{\max}^3}{\gamma q_{\min}^2} \|P_{k+1} - P^*(W_k, Q_k)\|^2 \tag{59}$$

$$\frac{2q_{\max}\gamma^2}{\alpha} \|Q_{k+1} - Q_k\|_{M_p(P_k)^\dagger}^2 \leq \frac{2q_{\max}\gamma^2}{\alpha q_{\min}^2} \|Q_{k+1} - Q_k\|^2. \quad (60)$$

By inequality (58), the last two terms in the right-hand side (RHS) of (51) is bounded by

$$\begin{aligned} & \frac{q_{\max}}{\alpha\gamma} \mathbb{E}_{b'_k} [\|W_{k+1} - W_k\|_{M_p(P_k)^\dagger}^2] + \frac{3L}{2} \mathbb{E}_{b'_k} [\|W_{k+1} - W_k\|^2] \\ & \leq \frac{q_{\max}}{\alpha q_{\min}^2 \gamma} \mathbb{E}_{b'_k} [\|W_{k+1} - W_k\|^2] + \frac{3L}{2} \mathbb{E}_{b'_k} [\|W_{k+1} - W_k\|^2] \\ & \stackrel{(a)}{\leq} \frac{2q_{\max}}{\alpha q_{\min}^2 \gamma} \mathbb{E}_{b'_k} [\|W_{k+1} - W_k\|^2] \leq \frac{2\beta^2 q_{\max}}{\alpha q_{\min}^2 \gamma} \|(P_{k+1} - Q_k) \odot F_w(W_k) - |P_{k+1} - Q_k| \odot G_w(W_k)\|^2 + \Theta\left(\frac{\beta q_{\max} \Delta w_{\min}}{\alpha q_{\min}^2 \gamma}\right) \\ & \leq \frac{4\beta^2 q_{\max}}{\alpha q_{\min}^2 \gamma} \|(P^*(W_k, Q_k) - Q_k) \odot F_w(W_k) - |P^*(W_k, Q_k) - Q_k| \odot G_w(W_k)\|^2 + \Theta\left(\frac{\beta q_{\max} \Delta w_{\min}}{\alpha q_{\min}^2 \gamma}\right) \\ & \stackrel{(b)}{\leq} + \frac{4\beta^2 q_{\max}}{\alpha q_{\min}^2 \gamma} \|(P_{k+1} - Q_k) \odot F_w(W_k) - |P_{k+1} - Q_k| \odot G_w(W_k) \\ & \quad - ((P^*(W_k, Q_k) - Q_k) \odot F_w(W_k) - |P^*(W_k, Q_k) - Q_k| \odot G_w(W_k))\|^2 + \Theta\left(\frac{\gamma \Delta w_{\min}}{q_{\max}^2}\right) \\ & \stackrel{(c)}{\leq} \frac{4\beta^2 q_{\max}}{\alpha q_{\min}^2 \gamma} \|(P^*(W_k, Q_k) - Q_k) \odot F_w(W_k) - |P^*(W_k, Q_k) - Q_k| \odot G_w(W_k)\|^2 + \frac{4\beta^2 q_{\max}^3}{\alpha q_{\min}^2 \gamma} \|P_{k+1} - P^*(W_k, Q_k)\|^2 \\ & \quad + \Theta\left(\frac{\gamma \Delta w_{\min}}{q_{\min}}\right) \end{aligned} \quad (61)$$

where (a) holds if the learning rate α is sufficiently small such that $\alpha \leq \frac{2q_{\max}}{3q_{\min}\gamma L}$; (b) is because and $\beta \leq \frac{\gamma}{2q_{\max}}$, $\alpha \leq \frac{q_{\max}}{q_{\min}^2 \gamma L}$; and (c) comes from the fact that the analog update is Lipschitz continuous (see Lemma A.2).

By inequality (60), the two terms related to $\|Q_{k+1} - Q_k\|^2$ in the RHS of (51) is bounded by

$$\frac{2q_{\max}\gamma}{\alpha} \mathbb{E}_{\xi_k} [\|Q_{k+1} - Q_k\|_{M_p(P_k)^\dagger}^2] + L\gamma^2 \mathbb{E}_{\xi_k} [\|Q_{k+1} - Q_k\|^2] \leq \frac{3q_{\max}\gamma}{\alpha q_{\min}^2} \mathbb{E}_{\xi_k} [\|Q_{k+1} - Q_k\|^2]. \quad (62)$$

where the last inequality holds when choosing stepsize $\alpha \leq \frac{q_{\max}}{q_{\min}^2 \gamma L}$.

With all the inequalities and lemmas above, we are ready to prove the main conclusion in Theorem 3.7 now. Define a Lyapunov function by

$$V_k := f(\bar{W}_k) - f^* + C_1 \|P^*(W_k, Q_k) - Q_k\|^2 + C_2 (\varphi(P_k) - \varphi^*). \quad (63)$$

By Lemmas E.1 and E.2, we show that V_k has sufficient descent in expectation

$$\begin{aligned} & \mathbb{E}_{\xi_k, b_k, b'_k} [V_{k+1}] = \mathbb{E}_{\xi_k, b_k, b'_k} [f(\bar{W}_{k+1}) - f^* + C_1 \|P^*(W_{k+1}, Q_{k+1}) - Q_{k+1}\|^2 + C_2 (\varphi(P_{k+1}) - \varphi^*)] \\ & \leq f(\bar{W}_k) - f^* - \frac{\alpha\gamma}{8q_{\max}} \|\nabla f(\bar{W}_k)\|_{M_p(P_k)}^2 + \frac{\alpha\sigma^2\gamma}{2} \left\| \frac{G_p(P_k)}{\sqrt{F_p(P_k)}} \right\|_\infty^2 + 3\alpha^2 L q_{\max}^2 \sigma^2 + \Theta\left(\frac{\gamma \Delta w_{\min}}{q_{\min}}\right) \\ & \quad + \frac{6q_{\max}\gamma^2 \eta^2}{\alpha q_{\min}^2} \mathbb{E}_{\xi_k, b_k} [\|P_{k+1} - P^*(W_k, Q_k)\|^2 + \|P^*(W_k, Q_k) - Q_k\|^2] + \frac{4\beta^2 q_{\max}^3}{\alpha q_{\min}^2} \mathbb{E}_{\xi_k, b_k} [\|P_{k+1} - P^*(W_k, Q_k)\|^2] \\ & \quad - \left(\frac{1}{2\alpha q_{\max}} - 3\gamma^2 L \right) \mathbb{E}_{\xi_k, b_k} [\|P_{k+1} - P_k + \alpha(\nabla f(\bar{W}_k; \xi_k) - \nabla f(\bar{W}_k)) \odot F_p(P_k) - b_k\|^2] \\ & \quad + \frac{4\beta^2 q_{\max}}{\alpha q_{\min}^2} \|(P^*(W_k, Q_k) - Q_k) \odot F_w(W_k) - |P^*(W_k, Q_k) - Q_k| \odot G_w(W_k)\|^2 \\ & \quad + C_1 \left(\|P^*(W_k, Q_k) - Q_k\|^2 - \frac{\beta}{2\gamma q_{\max}} \|P^*(W_k, Q_k) - Q_k\|_{M_w(W_k)}^2 + \frac{3\beta q_{\max}^3}{\gamma q_{\min}^2} \mathbb{E}_{\xi_k} [\|P_{k+1} - P^*(W_k, Q_k)\|^2] \right) \end{aligned} \quad (64)$$

$$\begin{aligned}
 & -\frac{\beta}{2\gamma q_{\max}} \|(P^*(W_k, Q_k) - Q_k) \odot F_w(W_k) - |P^*(W_k, Q_k) - Q_k| \odot G_w(W_k)\|^2 + \Theta\left(\frac{\Delta w_{\min}}{\gamma q_{\max}}\right) \\
 & + C_2 \left(\varphi(P_k) - \varphi^* - \frac{\alpha C_\star}{2} \|G_p(P_k)\|^2 + \alpha^2 L q_{\max}^2 \sigma^2 + \frac{\alpha q_{\max}^2}{2C_\star} \|\nabla f(\bar{W}_k)\|^2 \right. \\
 & \quad \left. + L \mathbb{E}_{\xi_k, b_k} [\|P_{k+1} - P_k + \alpha(\nabla f(\bar{W}_k; \xi_k) - \nabla f(\bar{W}_k)) \odot F_p(P_k) - b_k\|^2] + \Theta\left(\frac{\Delta w_{\min}}{\gamma q_{\max}}\right) \right) \\
 & \leq V_k - \left(\frac{\alpha q_{\min}^2}{8q_{\max}} - \frac{C_2 \alpha q_{\max}^2}{2C_\star} \right) \|\nabla f(\bar{W}_k)\|^2 - \left(\frac{\alpha C_2 C_\star}{2} - \frac{\alpha \sigma^2}{2q_{\min}} \right) \|G_p(P_k)\|^2 + \Theta\left(\frac{(C_1 + C_2)\Delta w_{\min}}{\gamma q_{\max}} + \frac{\gamma \Delta w_{\min}}{q_{\min}}\right) \\
 & \quad + (3 + C_2) \alpha^2 L q_{\max}^2 \sigma^2 - \left(\frac{\beta q_{\min}^2}{2\gamma q_{\max}} C_1 - \frac{6q_{\max} \gamma^2 \eta^2}{\alpha q_{\min}^2} \right) \|P^*(W_k, Q_k) - Q_k\|^2 \\
 & \quad - \left(\frac{\beta}{2\gamma q_{\max}} C_1 - \frac{4\beta^2 q_{\max}}{\alpha q_{\min}^2} \right) \|(P^*(W_k, Q_k) - Q_k) \odot F_w(W_k) - |P^*(W_k, Q_k) - Q_k| \odot G_w(W_k)\|^2 \\
 & \quad + \left(\frac{3\beta q_{\max}^3}{\gamma q_{\min}^2} C_1 + \frac{4\beta^2 q_{\max}^3}{\alpha q_{\min}^2} + \frac{6q_{\max} \gamma^2 \eta^2}{\alpha q_{\min}^2} \right) \mathbb{E}_{\xi_k} [\|P_{k+1} - P^*(W_k, Q_k)\|^2] \\
 & \quad - \left(\frac{1}{2\alpha q_{\max}} - 3\gamma^2 L - C_2 L \right) \mathbb{E}_{\xi_k, b_k} [\|P_{k+1} - P_k + \alpha(\nabla f(\bar{W}_k; \xi_k) - \nabla f(\bar{W}_k)) \odot F_p(P_k) - b_k\|^2].
 \end{aligned}$$

The first inequality holds for:

$$\begin{aligned}
 \mathbb{E}_{\xi_k} [\|Q_{k+1} - Q_k\|^2] &= \mathbb{E}_{\xi_k} [\|\eta(P_{k+1} - Q_k)\|^2] \\
 &\leq 2\eta^2 \mathbb{E}_{\xi_k} [\|P_{k+1} - P^*(W_k, Q_k)\|^2 + \|P^*(W_k, Q_k) - Q_k\|^2]
 \end{aligned} \tag{65}$$

and the learning rate β is sufficiently small such that $\frac{2\beta^2}{\gamma^2} \leq \frac{\beta q_{\max}^3}{\gamma q_{\min}^2}$. Let $C_1 = \frac{10\beta\gamma q_{\max}^2}{\alpha q_{\min}^2}$, which leads to the positive coefficient in front of $\|P^*(W_k, Q_k) \odot F_w(W_k) - |P^*(W_k, Q_k)| \odot G_w(W_k)\|^2$. Let α large enough, which leads to the positive coefficient in front of $\mathbb{E}_{\xi_k, b_k} [\|P_{k+1} - P_k + \alpha(\nabla f(\bar{W}_k; \xi_k) - \nabla f(\bar{W}_k)) \odot F_p(P_k) - b_k\|^2]$.

$$\begin{aligned}
 \mathbb{E}_{\xi_k, b_k} [V_{k+1}] &\leq V_k - \left(\frac{\alpha q_{\min}^2}{8q_{\max}} - \frac{C_2 \alpha q_{\max}^2}{2C_\star} \right) \|\nabla f(\bar{W}_k)\|^2 - \left(\frac{\alpha C_2 C_\star}{2} - \frac{\alpha \sigma^2}{2q_{\min}} \right) \|G_p(P_k)\|^2 \\
 &\quad + \left(\frac{30\beta^2 q_{\max}^5}{\alpha q_{\min}^4} + \frac{4\beta^2 q_{\max}^3}{\alpha q_{\min}^2} + \frac{6q_{\max} \gamma^2 \eta^2}{\alpha q_{\min}^2} \right) \mathbb{E}_{\xi_k, b_k} [\|P_{k+1} - P^*(W_k, Q_k)\|^2] \\
 &\quad - \left(\frac{\beta q_{\min}^2}{2\gamma q_{\max}} C_1 - \frac{6q_{\max} \gamma^2 \eta^2}{\alpha q_{\min}^2} \right) \|P^*(W_k, Q_k) - Q_k\|^2 + (3 + C_2) \alpha^2 L q_{\max}^2 \sigma^2 + \Theta(\Delta w_{\min}).
 \end{aligned} \tag{66}$$

Now we bound the term $\|\nabla f(\bar{W}_k)\|^2$ in (66) by μ -PL condition (which is implied by Assumption 3.3)

$$\frac{\alpha q_{\min}^2}{8q_{\max}} \|\nabla f(\bar{W}_k)\|^2 \geq \frac{\alpha \mu q_{\min}^2}{4q_{\max}} (f(\bar{W}_k) - f^*). \tag{67}$$

Notice that the $\|P_{k+1} - P^*(W_k, Q_k)\|^2$ appears in the RHS of (66) above, we also need the following lemma to bound it.

Lemma E.4 (Quadratic growth, Theorem 2, (Karimi et al., 2016)). *Under Assumption 3.3, defining the optimal solution as $W^* := \arg \min_W f(W)$, it holds that*

$$\frac{2}{\mu} (f(W) - f^*) \geq \|W - W^*\|^2. \tag{68}$$

Replacing W in (68) with $W_k + \gamma(P_{k+1} - Q_k)$, we bound the term $\|P_{k+1} - P^*(W_k, Q_k)\|^2$ in (53) by

$$\frac{2}{\mu} (f(W_k + \gamma(P_{k+1} - Q_k)) - f^*) \geq \|W_k + \gamma P_{k+1} - \gamma Q_k - W^*\|^2 = \gamma^2 \|P_{k+1} - P^*(W_k, Q_k)\|^2. \tag{69}$$

By inequality (52) in Lemma E.1 and (69), we bound the $\|P_{k+1} - P^*(W_k, Q_k)\|^2$ by

$$\begin{aligned}
 & \left(\frac{30\beta^2 q_{\max}^5}{\alpha q_{\min}^4} + \frac{4\beta^2 q_{\max}^3}{\alpha q_{\min}^2} + \frac{6q_{\max}\gamma^2\eta^2}{\alpha q_{\min}^2} \right) \mathbb{E}_{\xi_k, b_k} [\|P_{k+1} - P^*(W_k, Q_k)\|^2] \\
 & \stackrel{(a)}{\leq} \left(\frac{34\beta^2 q_{\max}^5}{\alpha q_{\min}^4} + \frac{6q_{\max}\gamma^2\eta^2}{\alpha q_{\min}^2} \right) \mathbb{E}_{\xi_k, b_k} [\|P_{k+1} - P^*(W_k, Q_k)\|^2] \\
 & \stackrel{(b)}{\leq} \left(\frac{68\beta^2 q_{\max}^5}{\alpha\gamma^2\mu q_{\min}^4} + \frac{12q_{\max}\eta^2}{\alpha q_{\min}^2\mu} \right) \mathbb{E}_{\xi_k, b_k} [f(W_k + \gamma(P_{k+1} - Q_k)) - f^*] \\
 & \stackrel{(c)}{\leq} \left(\frac{68\beta^2 q_{\max}^5}{\alpha\gamma^2\mu q_{\min}^4} + \frac{12q_{\max}\eta^2}{\alpha q_{\min}^2\mu} \right) \left(f(\bar{W}_k) - f^* + \frac{\alpha\sigma^2}{2} \left\| \frac{G_p(P_k)}{\sqrt{F_p(P_k)}} \right\|_{\infty}^2 + \alpha^2 L q_{\max}^2 \sigma^2 + \Theta\left(\frac{\gamma\Delta w_{\min}}{q_{\min}}\right) \right) \\
 & \quad - \left(\frac{68\beta^2 q_{\max}^5}{\alpha\gamma^2\mu q_{\min}^4} + \frac{12q_{\max}\eta^2}{\alpha q_{\min}^2\mu} \right) \frac{\alpha}{2q_{\max}} \|\nabla f(\bar{W}_k)\|^2 \\
 & \stackrel{(d)}{\leq} \left(\frac{68\beta^2 q_{\max}^5}{\alpha\gamma^2\mu q_{\min}^4} + \frac{12q_{\max}\eta^2}{\alpha q_{\min}^2\mu} \right) (f(\bar{W}_k) - f^*) + \mathcal{O}\left(\beta^2\sigma^2 \left\| \frac{G_p(P_k)}{\sqrt{F_p(P_k)}} \right\|_{\infty}^2 + \alpha\beta^2 q_{\max}^2 \sigma^2\right) + \Theta\left(\frac{\gamma q_{\max}\Delta w_{\min}}{q_{\min}^2}\right) \\
 & \stackrel{(e)}{\leq} \left(\frac{68\beta^2 q_{\max}^5}{\alpha\gamma^2\mu q_{\min}^4} + \frac{12q_{\max}\eta^2}{\alpha q_{\min}^2\mu} \right) (f(\bar{W}_k) - f^*) + \alpha\sigma^2 \left\| \frac{G_p(P_k)}{\sqrt{F_p(P_k)}} \right\|_{\infty}^2 + \alpha^2 L q_{\max}^2 \sigma^2 + \Theta\left(\frac{\gamma q_{\max}\Delta w_{\min}}{q_{\min}^2}\right)
 \end{aligned} \tag{70}$$

where (a) uses $\frac{q_{\max}}{q_{\min}} \geq 1$; (b) comes from (69); (c) comes from (52) in Lemma E.1; (d) holds by setting $\eta \leq \frac{\beta q_{\max}^2}{\gamma q_{\min}}$ and $\beta \leq \frac{\gamma}{q_{\max}}$; (e) holds given α and β is sufficiently small.

Substituting (67) and (70) back into (66) yields

$$\begin{aligned}
 & \mathbb{E}_{\xi_k, b_k} [V_{k+1}] \\
 & \leq V_k - \left(\frac{\alpha C_2 C_{\star}}{2} - \frac{3\alpha\sigma^2}{2q_{\min}} \right) \|G_p(P_k)\|^2 - \left(\frac{\alpha\mu q_{\min}^2}{4q_{\max}} - \frac{68\beta^2 q_{\max}^5}{\alpha\gamma^2\mu q_{\min}^4} - \frac{12q_{\max}\eta^2}{\alpha q_{\min}^2\mu} - \frac{C_2\alpha q_{\max}^2}{2C_{\star}} \right) (f(\bar{W}_k) - f^*) \\
 & \quad - \left(\frac{\beta q_{\min}^2}{2\gamma q_{\max}} C_1 - \frac{6q_{\max}\gamma^2\eta^2}{\alpha q_{\min}^2} \right) \|P^*(W_k, Q_k) - Q_k\|^2 + (4 + C_2)\alpha^2 L q_{\max}^2 \sigma^2 + \Theta(\Delta w_{\min}) \\
 & = V_k - \left(\frac{\alpha C_2 C_{\star}}{2} - \frac{3\alpha\sigma^2}{2q_{\min}} \right) \|G_p(P_k)\|^2 - \left(\frac{\alpha\mu q_{\min}^2}{8q_{\max}} - \frac{12q_{\max}\eta^2}{\alpha q_{\min}^2\mu} - \frac{C_2\alpha q_{\max}^2}{2C_{\star}} \right) (f(\bar{W}_k) - f^*) \\
 & \quad - \left(\frac{\alpha\mu q_{\min}^5}{8\sqrt{34}q_{\max}^4} - \frac{12\sqrt{34}q_{\max}^2\eta^2}{5\alpha\mu q_{\min}^3} \right) C_1 \|P^*(W_k, Q_k) - Q_k\|^2 + (4 + C_2)\alpha^2 L q_{\max}^2 \sigma^2 + \Theta(\Delta w_{\min}) \\
 & \leq V_k - \left(\frac{\alpha\mu q_{\min}^5}{8\sqrt{34}q_{\max}^4} - \frac{12\sqrt{34}q_{\max}^2\eta^2}{5\alpha\mu q_{\min}^3} \right) ((f(\bar{W}_k) - f^*) + C_1 \|P^*(W_k, Q_k) - Q_k\|^2 + \|G_p(P_k)\|^2) \\
 & \quad + (4 + C_2)\alpha^2 L q_{\max}^2 \sigma^2 + \Theta(\Delta w_{\min}) \\
 & \leq V_k - \frac{\alpha\mu q_{\min}^5}{16\sqrt{34}q_{\max}^4} ((f(\bar{W}_k) - f^*) + C_1 \|P^*(W_k, Q_k) - Q_k\|^2 + \|G_p(P_k)\|^2) + 5\alpha^2 L q_{\max}^2 \sigma^2 + \Theta(\Delta w_{\min})
 \end{aligned} \tag{71}$$

where the second step chooses the learning rate by $\beta = \frac{\alpha\gamma\mu q_{\min}^3}{4\sqrt{34}q_{\max}^3}$. The third step holds since $\frac{q_{\max}^2}{q_{\min}^2} \geq 1$, and choose $\frac{4\sigma^2}{C_{\star} q_{\min}} \leq C_2 \leq \frac{\mu q_{\min}^2 C_{\star}}{8q_{\max}^3}$, such that $\frac{\alpha C_2 C_{\star}}{2} - \frac{3\alpha\sigma^2}{2q_{\min}} \geq \frac{\alpha\mu q_{\min}^5}{8\sqrt{34}q_{\max}^4} - \frac{12\sqrt{34}q_{\max}^2\eta^2}{5\alpha\mu q_{\min}^3}$ and $\frac{C_2\alpha q_{\max}^2}{2C_{\star}} \leq \frac{\alpha\mu q_{\min}^2}{16q_{\max}}$. The last step chooses $\eta \leq \frac{\alpha\mu q_{\min}^4}{52q_{\max}^3}$ such that $\frac{12\sqrt{34}q_{\max}^2\eta^2}{5\alpha\mu q_{\min}^3} \leq \frac{\alpha\mu q_{\min}^5}{32\sqrt{34}q_{\max}^4}$.

Rearranging (71), taking expectations with respect to $\{\xi_k\}_{k=0}^K$, and averaging over $k = 0, \dots, K$, and choosing the stepsize $\alpha = \Theta(K^{-1/2})$, we obtain

$$\frac{1}{K} \sum_{k=0}^K \mathbb{E} \left((f(\bar{W}_k) - f^*) + C_1 \|P^*(W_k, Q_k) - Q_k\|^2 + \|G_p(P_k)\|^2 \right)$$

$$\leq \frac{16\sqrt{34}q_{\max}^4}{\mu q_{\min}^5} \left(\frac{V_0 - V_{[K]}}{\alpha K} + 5\alpha L q_{\max}^2 \sigma^2 \right) + \Theta(\Delta w_{\min}) \leq \mathcal{O}\left(\frac{\kappa_2^5}{\sqrt{K}}\right) + \Theta(\Delta w_{\min}). \quad (72)$$

Since $\|P^*(W_k, Q_k) - Q_k\| = \|W_k - W^*\|/\gamma$ and $f(\bar{W}_k) - f^* \geq \frac{\mu}{2}\|\bar{W}_k - W^*\|^2 = \frac{\mu}{2}\|W_k - W^* + \gamma(P_k - Q_k)\|^2$, we know that

$$\begin{aligned} \|P_k - Q_k\|^2 &\leq 2\gamma^{-2}\|\bar{W}_k - W^*\|^2 + 2\gamma^{-2}\|W_k - W^*\|^2 \leq 2\gamma^{-2}\|\bar{W}_k - W^*\|^2 + 2\|P^*(W_k, Q_k) - Q_k\|^2 \\ &\leq \frac{4}{\mu}\gamma^{-2}(f(\bar{W}_k) - f^*) + 2\|P^*(W_k, Q_k) - Q_k\|^2 \end{aligned}$$

and thus by choosing $C_1 = \mathcal{O}(\gamma^2)$, we have

$$\begin{aligned} &\frac{1}{K} \sum_{k=0}^K \mathbb{E} \left(\|W_k - W^*\|^2 + C_1 \|P_k - Q_k\|^2 + \|G_p(P_k)\|^2 \right) \\ &\leq \frac{16\sqrt{34}q_{\max}^4}{\mu q_{\min}^5} \left(\frac{V_0 - V_{[K]}}{\alpha K} + 5\alpha L q_{\max}^2 \sigma^2 \right) + \Theta(\Delta w_{\min}) \leq \mathcal{O}\left(\frac{\kappa_2^5}{\kappa_1 \sqrt{K}}\right) + \Theta(\Delta w_{\min}). \end{aligned} \quad (73)$$

The proof of Theorem 3.7 is completed. $\square$

E.2. Proof of Lemma E.1: Descent of sequence $\bar{W}_k$

Lemma E.1 (Descent lemma of lower-level problem). *Under Assumptions 3.1-3.2, it holds that*

$$\begin{aligned} \mathbb{E}_{\xi_k, b_k} [f(\bar{W}_{k+1})] &\leq f(\bar{W}_k) - \frac{\alpha\gamma}{8q_{\max}} \|\nabla f(\bar{W}_k)\|_{M_p(P_k)}^2 + \frac{\alpha\gamma\sigma^2}{2} \left\| \frac{G_p(P_k)}{\sqrt{F_p(P_k)}} \right\|_{\infty}^2 + \Theta\left(\frac{\gamma\Delta w_{\min}}{q_{\min}}\right) \\ &\quad + \frac{2q_{\max}\gamma}{\alpha} \mathbb{E}_{\xi_k} [\|Q_{k+1} - Q_k\|_{M_p(P_k)}^2] + \frac{3\gamma^2 L}{2} \mathbb{E}_{\xi_k} [\|Q_{k+1} - Q_k\|^2] \\ &\quad - \left(\frac{\gamma}{2\alpha q_{\max}} - 3\gamma^2 L \right) \mathbb{E}_{\xi_k, b_k} [\|P_{k+1} - P_k + \alpha(\nabla f(\bar{W}_k; \xi_k) - \nabla f(\bar{W}_k)) \odot F_p(P_k) - b_k\|^2] \\ &\quad + 3\alpha^2 \gamma^2 L q_{\max}^2 \sigma^2 + \frac{q_{\max}}{\alpha\gamma} \mathbb{E}_{\xi_k} [\|W_{k+1} - W_k\|_{M_p(P_k)}^2] + \frac{3L}{2} \mathbb{E}_{\xi_k} [\|W_{k+1} - W_k\|^2]. \end{aligned} \quad (51)$$

Keeping W_k fixed, we also have

$$\begin{aligned} &\mathbb{E}_{\xi_k, b_k} [f(W_k + \gamma P_{k+1} - \gamma Q_k)] \\ &\leq f(\bar{W}_k) - \frac{\alpha\gamma}{2q_{\max}} \|\nabla f(\bar{W}_k)\|_{M_p(P_k)}^2 + \frac{\alpha\gamma\sigma^2}{2} \left\| \frac{G_p(P_k)}{\sqrt{F_p(P_k)}} \right\|_{\infty}^2 + \Theta\left(\frac{\gamma\Delta w_{\min}}{q_{\min}}\right) \\ &\quad - \left(\frac{\gamma}{2\alpha q_{\max}} - \gamma^2 L \right) \mathbb{E}_{\xi_k, b_k} [\|P_{k+1} - P_k + \alpha(\nabla f(\bar{W}_k; \xi_k) - \nabla f(\bar{W}_k)) \odot F_p(P_k) - b_k\|^2] + \alpha^2 \gamma^2 L q_{\max}^2 \sigma^2. \\ &\leq f(\bar{W}_k) - \frac{\alpha\gamma}{2q_{\max}} \|\nabla f(\bar{W}_k)\|_{M_p(P_k)}^2 + \frac{\alpha\gamma\sigma^2}{2} \left\| \frac{G_p(P_k)}{\sqrt{F_p(P_k)}} \right\|_{\infty}^2 + \alpha^2 \gamma^2 L q_{\max}^2 \sigma^2 + \Theta\left(\frac{\gamma\Delta w_{\min}}{q_{\min}}\right) \end{aligned} \quad (52)$$

Lemma E.1. The L -smooth assumption (Assumption 3.1) implies that

$$\begin{aligned} \mathbb{E}_{\xi_k, b_k} [f(\bar{W}_{k+1})] &\leq f(\bar{W}_k) + \mathbb{E}_{\xi_k, b_k} [\langle \nabla f(\bar{W}_k), \bar{W}_{k+1} - \bar{W}_k \rangle] + \frac{L}{2} \mathbb{E}_{\xi_k, b_k} [\|\bar{W}_{k+1} - \bar{W}_k\|^2] \\ &= f(\bar{W}_k) + \gamma \underbrace{\mathbb{E}_{\xi_k, b_k} [\langle \nabla f(\bar{W}_k), P_{k+1} - P_k \rangle]}_{(a)} + \underbrace{\mathbb{E}_{\xi_k} [\langle \nabla f(\bar{W}_k), W_{k+1} - W_k \rangle]}_{(b)} + \underbrace{\frac{L}{2} \mathbb{E}_{\xi_k, b_k} [\|\bar{W}_{k+1} - \bar{W}_k\|^2]}_{(c)} \\ &\quad - \gamma \underbrace{\mathbb{E}_{\xi_k} [\langle \nabla f(\bar{W}_k), Q_{k+1} - Q_k \rangle]}_{(d)} \end{aligned} \quad (74)$$

Next, we will handle each term in the RHS of (74) separately.

Bound of the second term (a). To bound term (a) in the RHS of (74), we leverage the assumption that noise has expectation 0 (Assumption 3.2)

$$\begin{aligned}
 & \mathbb{E}_{\xi_k, b_k} [\langle \nabla f(\bar{W}_k), P_{k+1} - P_k \rangle] \\
 &= \alpha \mathbb{E}_{\xi_k, b_k} \left[\left\langle \nabla f(\bar{W}_k) \odot \sqrt{F_p(P_k)}, \frac{P_{k+1} - P_k}{\alpha \sqrt{F_p(P_k)}} + (\nabla f(\bar{W}_k; \xi_k) - \nabla f(\bar{W}_k)) \odot \sqrt{F_p(P_k)} - \frac{b_k}{\alpha \sqrt{F_p(P_k)}} \right\rangle \right] \\
 &= -\frac{\alpha}{2} \|\nabla f(\bar{W}_k) \odot \sqrt{F_p(P_k)}\|^2 \\
 &\quad - \frac{1}{2\alpha} \mathbb{E}_{\xi_k, b_k} \left[\left\| \frac{P_{k+1} - P_k}{\sqrt{F_p(P_k)}} + \alpha (\nabla f(\bar{W}_k; \xi_k) - \nabla f(\bar{W}_k)) \odot \sqrt{F_p(P_k)} - \frac{b_k}{\sqrt{F_p(P_k)}} \right\|^2 \right] \\
 &\quad + \frac{1}{2\alpha} \mathbb{E}_{\xi_k, b_k} \left[\left\| \frac{P_{k+1} - P_k}{\sqrt{F_p(P_k)}} + \alpha \nabla f(\bar{W}_k; \xi_k) \odot \sqrt{F_p(P_k)} - \frac{b_k}{\sqrt{F_p(P_k)}} \right\|^2 \right].
 \end{aligned} \tag{75}$$

The second term in the RHS of (75) can be bounded by

$$\begin{aligned}
 & \frac{1}{2\alpha} \mathbb{E}_{\xi_k, b_k} \left[\left\| \frac{P_{k+1} - P_k}{\sqrt{F_p(P_k)}} + \alpha (\nabla f(\bar{W}_k; \xi_k) - \nabla f(\bar{W}_k)) \odot \sqrt{F_p(P_k)} - \frac{b_k}{\sqrt{F_p(P_k)}} \right\|^2 \right] \\
 &= \frac{1}{2\alpha} \mathbb{E}_{\xi_k, b_k} \left[\left\| \frac{P_{k+1} - P_k + \alpha (\nabla f(\bar{W}_k; \xi_k) - \nabla f(\bar{W}_k)) \odot F_p(P_k) - b_k}{\sqrt{F_p(P_k)}} \right\|^2 \right] \\
 &\geq \frac{1}{2\alpha q_{\max}} \mathbb{E}_{\xi_k, b_k} [\|P_{k+1} - P_k + \alpha (\nabla f(\bar{W}_k; \xi_k) - \nabla f(\bar{W}_k)) \odot F_p(P_k) - b_k\|^2].
 \end{aligned} \tag{76}$$

The third term in the RHS of (75) can be bounded by the bounded variance (Assumption 3.2)

$$\frac{1}{2\alpha} \mathbb{E}_{\xi_k, b_k} \left[\left\| \frac{P_{k+1} - P_k}{\sqrt{F_p(P_k)}} + \alpha \nabla f(\bar{W}_k; \xi_k) \odot \sqrt{F_p(P_k)} - \frac{b_k}{\sqrt{F_p(P_k)}} \right\|^2 \right] \tag{77}$$

$$\leq \frac{\alpha}{2} \mathbb{E}_{\xi_k, b_k} \left[\left\| |\nabla f(\bar{W}_k; \xi_k)| \odot \frac{G_p(P_k)}{\sqrt{F_p(P_k)}} - \frac{b_k}{\alpha \sqrt{F_p(P_k)}} \right\|^2 \right] \tag{78}$$

$$\stackrel{(a)}{=} \frac{\alpha}{2} \mathbb{E}_{\xi_k, b_k} \left[\left\| \nabla f(\bar{W}_k; \xi_k) \odot \frac{G_p(P_k)}{\sqrt{F_p(P_k)}} - \frac{b_k}{\alpha \sqrt{F_p(P_k)}} \right\|^2 \right] \tag{79}$$

$$\stackrel{(b)}{\leq} \frac{\alpha}{2} \left\| \nabla f(\bar{W}_k) \odot \frac{G_p(P_k)}{\sqrt{F_p(P_k)}} \right\|^2 + \frac{\alpha \sigma^2}{2} \left\| \frac{G_p(P_k)}{\sqrt{F_p(P_k)}} \right\|_{\infty}^2 + \Theta \left(\frac{\Delta w_{\min}}{q_{\min}} \right).$$

where (a) is because $\mathbb{E}_{b_k} \left[\left\langle |\nabla f(\bar{W}_k; \xi_k)| \odot \frac{G_p(P_k)}{\sqrt{F_p(P_k)}}, \frac{b_k}{\alpha \sqrt{F_p(P_k)}} \right\rangle \right] = 0$ according to Assumption 3.4 and (b) comes from $\mathbb{E}[\|A\|^2] = \|\mathbb{E}[A]\|^2 + \mathbb{E}[\|A - \mathbb{E}[A]\|^2]$ and Assumption 3.2.

Notice that the first term in the RHS of (75) and the second term in the RHS of (77) can be bounded together

$$\begin{aligned}
 & -\frac{\alpha}{2} \|\nabla f(\bar{W}_k) \odot \sqrt{F_p(P_k)}\|^2 + \frac{\alpha}{2} \left\| \nabla f(\bar{W}_k) \odot \frac{G_p(P_k)}{\sqrt{F_p(P_k)}} \right\|^2 \\
 &= -\frac{\alpha}{2} \sum_{d \in [D]} \left([\nabla f(\bar{W}_k)]_d^2 \left([F_p(P_k)]_d - \frac{[G_p(P_k)]_d^2}{[F_p(P_k)]_d} \right) \right)
 \end{aligned} \tag{80}$$

$$\begin{aligned}
 &= -\frac{\alpha}{2} \sum_{d \in [D]} \left([\nabla f(\bar{W}_k)]_d^2 \left(\frac{[F_p(P_k)]_d^2 - [G_p(P_k)]_d^2}{[F_p(P_k)]_d} \right) \right) \\
 &\leq -\frac{\alpha}{2q_{\max}} \sum_{d \in [D]} ([\nabla f(\bar{W}_k)]_d^2 ([F_p(P_k)]_d^2 - [G_p(P_k)]_d^2)) \\
 &= -\frac{\alpha}{2q_{\max}} \|\nabla f(\bar{W}_k)\|_{M_p(P_k)}^2 \leq 0.
 \end{aligned}$$

Plugging (76) to (80) into (75), we bound the term (a) by

$$\begin{aligned}
 \gamma \mathbb{E}_{\xi_k, b_k} [\langle \nabla f(\bar{W}_k), P_{k+1} - P_k \rangle] &\leq -\frac{\alpha\gamma}{2q_{\max}} \|\nabla f(\bar{W}_k)\|_{M_p(P_k)}^2 + \frac{\alpha\gamma\sigma^2}{2} \left\| \frac{G_p(P_k)}{\sqrt{F_p(P_k)}} \right\|_{\infty}^2 + \Theta \left(\frac{\gamma\Delta w_{\min}}{q_{\min}} \right) \\
 &\quad - \frac{\gamma}{2\alpha q_{\max}} \mathbb{E}_{\xi_k} \left[\|P_{k+1} - P_k + \alpha(\nabla f(\bar{W}_k; \xi_k) - \nabla f(\bar{W}_k)) \odot F_p(P_k) - b_k\|^2 \right].
 \end{aligned} \tag{81}$$

Bound of the third term (b). By Young's inequality, we have

$$\mathbb{E}_{\xi_k} [\langle \nabla f(\bar{W}_k), W_{k+1} - W_k \rangle] \leq \frac{\alpha\gamma}{4q_{\max}} \|\nabla f(\bar{W}_k)\|_{M_p(P_k)}^2 + \frac{q_{\max}}{\alpha\gamma} \mathbb{E}_{\xi_k} [\|W_{k+1} - W_k\|_{M_p(P_k)}^2]. \tag{82}$$

Bound of the fourth term (c). Repeatedly applying inequality $\|U + V\|^2 \leq 2\|U\|^2 + 2\|V\|^2$ for any $U, V \in \mathbb{R}^D$, we have

$$\begin{aligned}
 &\frac{L}{2} \mathbb{E}_{\xi_k, b_k} [\|\bar{W}_{k+1} - \bar{W}_k\|^2] \\
 &\leq \frac{3L}{2} \mathbb{E}_{\xi_k} [\|W_{k+1} - W_k\|^2] + \frac{3\gamma^2 L}{2} \mathbb{E}_{\xi_k, b_k} [\|P_{k+1} - P_k\|^2] + \frac{3\gamma^2 L}{2} \mathbb{E}_{\xi_k} [\|Q_{k+1} - Q_k\|^2] \\
 &\leq \frac{3L}{2} \mathbb{E}_{\xi_k} [\|W_{k+1} - W_k\|^2] + 3\gamma^2 L \mathbb{E}_{\xi_k, b_k} \left[\|P_{k+1} - P_k + \alpha(\nabla f(\bar{W}_k; \xi_k) - \nabla f(\bar{W}_k)) \odot F_p(P_k)\|^2 \right] \\
 &\quad + 3\alpha^2 L \gamma^2 \mathbb{E}_{\xi_k} \left[\|(\nabla f(\bar{W}_k; \xi_k) - \nabla f(\bar{W}_k)) \odot F_p(P_k)\|^2 \right] + \frac{3\gamma^2 L}{2} \mathbb{E}_{\xi_k} [\|Q_{k+1} - Q_k\|^2] \\
 &\leq \frac{3L}{2} \mathbb{E}_{\xi_k} [\|W_{k+1} - W_k\|^2] + 3\gamma^2 L \mathbb{E}_{\xi_k, b_k} \left[\|P_{k+1} - P_k + \alpha(\nabla f(\bar{W}_k; \xi_k) - \nabla f(\bar{W}_k)) \odot F_p(P_k) - b_k\|^2 \right] + 3\alpha^2 \gamma^2 L q_{\max}^2 \sigma^2 \\
 &\quad + \frac{3\gamma^2 L}{2} \mathbb{E}_{\xi_k} [\|Q_{k+1} - Q_k\|^2] + \Theta(3\gamma^2 L \alpha \Delta w_{\min})
 \end{aligned} \tag{83}$$

where the last inequality comes from Assumption 3.4 and the bounded variance assumption (Assumption 3.2)

$$\begin{aligned}
 &3\alpha^2 \gamma^2 L \mathbb{E}_{\xi_k} \left[\|(\nabla f(\bar{W}_k; \xi_k) - \nabla f(\bar{W}_k)) \odot F_p(P_k)\|^2 \right] \\
 &\leq 3\alpha^2 \gamma^2 L q_{\max}^2 \mathbb{E}_{\xi_k} \left[\|\nabla f(\bar{W}_k; \xi_k) - \nabla f(\bar{W}_k)\|^2 \right] \leq 3\alpha^2 \gamma^2 L q_{\max}^2 \sigma^2.
 \end{aligned} \tag{84}$$

Bound of the fifth term (d). By Young's inequality, we have

$$-\gamma \mathbb{E}_{\xi_k, b_k} [\langle \nabla f(\bar{W}_k), Q_{k+1} - Q_k \rangle] \leq \frac{\alpha\gamma}{8q_{\max}} \|\nabla f(\bar{W}_k)\|_{M_p(P_k)}^2 + \frac{2q_{\max}\gamma}{\alpha} \mathbb{E}_{\xi_k} [\|Q_{k+1} - Q_k\|_{M_p(P_k)}^2]. \tag{85}$$

Combination of the upper bound (a), (b), (c) and (d). Plugging (81), (82), (83) into (74) and letting $\alpha \leq \frac{1}{2\gamma L q_{\max}}$,

$$\begin{aligned}
 \mathbb{E}_{\xi_k, b_k} [f(\bar{W}_{k+1})] &\leq f(\bar{W}_k) - \frac{\alpha\gamma}{8q_{\max}} \|\nabla f(\bar{W}_k)\|_{M_p(P_k)}^2 + \frac{\alpha\gamma\sigma^2}{2} \left\| \frac{G_p(P_k)}{\sqrt{F_p(P_k)}} \right\|_{\infty}^2 + \Theta \left(\frac{\gamma\Delta w_{\min}}{q_{\min}} \right) \\
 &\quad + \frac{2q_{\max}\gamma}{\alpha} \mathbb{E}_{\xi_k} [\|Q_{k+1} - Q_k\|_{M_p(P_k)}^2] + \frac{3\gamma^2 L}{2} \mathbb{E}_{\xi_k} [\|Q_{k+1} - Q_k\|^2] \\
 &\quad - \left(\frac{\gamma}{2\alpha q_{\max}} - 3\gamma^2 L \right) \mathbb{E}_{\xi_k, b_k} \left[\|P_{k+1} - P_k + \alpha(\nabla f(\bar{W}_k; \xi_k) - \nabla f(\bar{W}_k)) \odot F_p(P_k) - b_k\|^2 \right]
 \end{aligned} \tag{86}$$

$$+ 3\alpha^2\gamma^2 Lq_{\max}^2\sigma^2 + \frac{q_{\max}}{\alpha\gamma} \mathbb{E}_{\xi_k} \left[\|W_{k+1} - W_k\|_{M_p(P_k)}^2 \right] + \frac{3L}{2} \mathbb{E}_{\xi_k} [\|W_{k+1} - W_k\|^2].$$

Now the proof of (51) is completed. Leveraging the L -smooth assumption (Assumption 3.1), we have

$$\mathbb{E}_{\xi_k, b_k} [f(W_k + \gamma(P_{k+1} - Q_k))] = f(\bar{W}_k) + \gamma \mathbb{E}_{\xi_k, b_k} [\langle \nabla f(\bar{W}_k), P_{k+1} - P_k \rangle] + \frac{L\gamma^2}{2} \mathbb{E}_{\xi_k, b_k} [\|P_{k+1} - P_k\|^2]. \quad (87)$$

Plugging (81) and (83) into (87), we have

$$\begin{aligned} & \mathbb{E}_{\xi_k, b_k} [f(W_k + \gamma P_{k+1} - \gamma Q_k)] \\ & \leq f(\bar{W}_k) - \frac{\alpha\gamma}{2q_{\max}} \|\nabla f(\bar{W}_k)\|_{M_p(P_k)}^2 + \frac{\alpha\gamma\sigma^2}{2} \left\| \frac{G_p(P_k)}{\sqrt{F_p(P_k)}} \right\|_{\infty}^2 + \Theta \left(\frac{\gamma\Delta w_{\min}}{q_{\min}} \right) \\ & \quad - \left(\frac{\gamma}{2\alpha q_{\max}} - \gamma^2 L \right) \mathbb{E}_{\xi_k, b_k} \left[\|P_{k+1} - P_k + \alpha(\nabla f(\bar{W}_k; \xi_k) - \nabla f(\bar{W}_k)) \odot F_p(P_k) - b_k\|^2 \right] + \alpha^2\gamma^2 Lq_{\max}^2\sigma^2. \\ & \leq f(\bar{W}_k) - \frac{\alpha\gamma}{2q_{\max}} \|\nabla f(\bar{W}_k)\|_{M_p(P_k)}^2 + \frac{\alpha\gamma\sigma^2}{2} \left\| \frac{G_p(P_k)}{\sqrt{F_p(P_k)}} \right\|_{\infty}^2 + \alpha^2\gamma^2 Lq_{\max}^2\sigma^2 + \Theta \left(\frac{\gamma\Delta w_{\min}}{q_{\min}} \right) \end{aligned} \quad (88)$$

where the second inequality holds by $\alpha \leq \frac{1}{2\gamma Lq_{\max}}$. Now the proof of (52) is completed. $\square$

E.3. Proof of Lemma E.2: Descent of sequence W_k

Lemma E.2 (Descent lemma of upper-level problem). *Under Assumption 2.1, it holds that*

$$\begin{aligned} \mathbb{E}_{b'_k} [\|P^*(W_{k+1}, Q_{k+1}) - Q_{k+1}\|^2] & \leq \|P^*(W_k, Q_k) - Q_k\|^2 - \frac{\beta}{2\gamma q_{\max}} \|P^*(W_k, Q_k) - Q_k\|_{M_w(W_k)}^2 + \Theta \left(\frac{\Delta w_{\min}}{\gamma q_{\max}} \right) \\ & \quad - \frac{\beta}{2\gamma q_{\max}} \|(P^*(W_k, Q_k) - Q_k) \odot F_w(W_k) - |P^*(W_k, Q_k) - Q_k| \odot G_w(W_k)\|^2 \\ & \quad + \frac{2\beta q_{\max}^3}{\gamma} \|P_{k+1} - P^*(W_k, Q_k)\|_{M_w(W_k)}^2 + \frac{2\beta^2}{\gamma^2} \|P_{k+1} - P^*(W_k, Q_k)\|^2. \end{aligned} \quad (53)$$

Lemma E.2. Recall the definition $P^*(W, Q) - Q = (W^* - W)/\gamma$, it holds that

$$\begin{aligned} \|P^*(W_{k+1}, Q_{k+1}) - Q_{k+1}\|^2 & = \frac{1}{\gamma^2} \|W_{k+1} - \text{Proj}_{\mathcal{W}^*}(W_{k+1})\|^2 \leq \frac{1}{\gamma^2} \|W_{k+1} - \text{Proj}_{\mathcal{W}^*}(W_k)\|^2 \\ & = \frac{1}{\gamma^2} \|W_k - \text{Proj}_{\mathcal{W}^*}(W_k)\|^2 + \frac{2}{\gamma^2} \langle W_k - \text{Proj}_{\mathcal{W}^*}(W_k), W_{k+1} - W_k \rangle + \frac{1}{\gamma^2} \|W_{k+1} - W_k\|^2 \\ & = \|P^*(W_k, Q_k) - Q_k\|^2 - \frac{2}{\gamma} \langle P^*(W_k, Q_k) - Q_k, W_{k+1} - W_k \rangle + \frac{1}{\gamma^2} \|W_{k+1} - W_k\|^2 \end{aligned} \quad (89)$$

where the first inequality comes from the fact that $\text{Proj}_{\mathcal{W}^*}(W_{k+1})$ is the closest point to W_{k+1} in $\mathcal{W}^*$. We bound the second term in the RHS of (89) by

$$\begin{aligned} & - \mathbb{E}_{b'_k} \left[\frac{2}{\gamma} \langle P^*(W_k, Q_k) - Q_k, W_{k+1} - W_k \rangle \right] \\ & = -\frac{2}{\gamma} \langle P^*(W_k, Q_k) - Q_k, \beta(P_{k+1} - Q_k) \odot F_w(W_k) - \beta|P_{k+1} - Q_k| \odot G_w(W_k) \rangle \\ & = -\frac{2\beta}{\gamma} \langle P^*(W_k, Q_k) - Q_k, (P^*(W_k, Q_k) - Q_k) \odot F_w(W_k) - |P^*(W_k, Q_k) - Q_k| \odot G_w(W_k) \rangle \\ & \quad - \frac{2\beta}{\gamma} \langle P^*(W_k, Q_k) - Q_k, (P_{k+1} - Q_k) \odot F_w(W_k) - |P_{k+1} - Q_k| \odot G_w(W_k) \rangle \\ & \quad - \langle (P^*(W_k, Q_k) - Q_k) \odot F_w(W_k) - |P^*(W_k, Q_k) - Q_k| \odot G_w(W_k) \rangle. \end{aligned} \quad (90)$$

where the first equality holds because $\mathbb{E}[b'_k] = 0$.

The first term in the RHS of (90) is bounded by

$$\begin{aligned}
 & -\frac{2\beta}{\gamma} \langle P^*(W_k, Q_k) - Q_k, (P^*(W_k, Q_k) - Q_k) \odot F_w(W_k) - |P^*(W_k, Q_k) - Q_k| \odot G_w(W_k) \rangle \\
 & = -\frac{2\beta}{\gamma} \left\langle (P^*(W_k, Q_k) - Q_k) \odot \sqrt{F_w(W_k)}, \frac{(P^*(W_k, Q_k) - Q_k) \odot F_w(W_k) - |P^*(W_k, Q_k) - Q_k| \odot G_w(W_k)}{\sqrt{F_w(W_k)}} \right\rangle \\
 & \stackrel{(a)}{=} -\frac{\beta}{\gamma} \|(P^*(W_k, Q_k) - Q_k) \odot \sqrt{F_w(W_k)}\|^2 + \frac{\beta}{\gamma} \left\| |P^*(W_k, Q_k) - Q_k| \odot \frac{G_w(W_k)}{\sqrt{F_w(W_k)}} \right\|^2 \\
 & \quad - \frac{\beta}{\gamma} \left\| (P^*(W_k, Q_k) - Q_k) \odot \sqrt{F_w(W_k)} + |P^*(W_k, Q_k) - Q_k| \odot \frac{G_w(W_k)}{\sqrt{F_w(W_k)}} \right\|^2 \\
 & \stackrel{(b)}{\leq} -\frac{\beta}{\gamma q_{\max}} \|P^*(W_k, Q_k) - Q_k\|_{M_w(W_k)}^2 - \frac{\beta}{\gamma} \left\| (P^*(W_k, Q_k) - Q_k) \odot \sqrt{F_w(W_k)} + |P^*(W_k, Q_k) - Q_k| \odot \frac{G_w(W_k)}{\sqrt{F_w(W_k)}} \right\|^2 \\
 & \leq -\frac{\beta}{\gamma q_{\max}} \|P^*(W_k, Q_k) - Q_k\|_{M_w(W_k)}^2 - \frac{\beta}{\gamma q_{\max}} \|(P^*(W_k, Q_k) - Q_k) \odot F_w(W_k) - |P^*(W_k, Q_k) - Q_k| \odot G_w(W_k)\|^2
 \end{aligned} \tag{91}$$

where (a) leverages $2\langle U, V \rangle = \|U\|^2 - \|V\|^2 - \|U - V\|^2$ for any $U, V \in \mathbb{R}^D$, (b) is achieved by

$$\begin{aligned}
 & -\frac{\beta}{2} \|(P^*(W_k, Q_k) - Q_k) \odot \sqrt{F_w(W_k)}\|^2 + \frac{\beta}{2} \left\| |P^*(W_k, Q_k) - Q_k| \odot \frac{G_w(W_k)}{\sqrt{F_w(W_k)}} \right\|^2 \\
 & = -\frac{\beta}{2} \sum_{d \in [D]} \left([P^*(W_k, Q_k) - Q_k]_d^2 \left([F_w(W_k)]_d - \frac{[G_w(W_k)]_d^2}{[F_w(W_k)]_d} \right) \right) \\
 & = -\frac{\beta}{2} \sum_{d \in [D]} \left([P^*(W_k, Q_k) - Q_k]_d^2 \left(\frac{[F_w(W_k)]_d^2 - [G_w(W_k)]_d^2}{[F_w(W_k)]_d} \right) \right) \\
 & \leq -\frac{\beta}{2q_{\max}} \sum_{d \in [D]} ([P^*(W_k, Q_k) - Q_k]_d^2 ([F_w(W_k)]_d^2 - [G_w(W_k)]_d^2)) = -\frac{\beta}{2q_{\max}} \|P^*(W_k, Q_k) - Q_k\|_{M_w(W_k)}^2.
 \end{aligned} \tag{92}$$

The second term in the RHS of (90) follows the Lipschitz continuity of analog update (see Lemma A.2)

$$\begin{aligned}
 & -\frac{2\beta}{\gamma} \langle P^*(W_k, Q_k) - Q_k, (P_{k+1} - Q_k) \odot F_w(W_k) - |P_{k+1} - Q_k| \odot G_w(W_k) \rangle \\
 & \quad - \langle (P^*(W_k, Q_k) - Q_k) \odot F_w(W_k) - |P^*(W_k, Q_k) - Q_k| \odot G_w(W_k) \rangle \\
 & = \frac{2\beta}{\gamma} \langle P^*(W_k, Q_k) - Q_k, -(P_{k+1} - Q_k) \odot F_w(W_k) + |P_{k+1} - Q_k| \odot G_w(W_k) \rangle \\
 & \quad + \langle (P^*(W_k, Q_k) - Q_k) \odot F_w(W_k) - |P^*(W_k, Q_k) - Q_k| \odot G_w(W_k) \rangle \\
 & \leq \frac{\beta}{2\gamma q_{\max}} \|P^*(W_k, Q_k) - Q_k\|_{M_w(W_k)}^2 + \frac{2\beta q_{\max}}{\gamma} \times \left\| (P_{k+1} - Q_k) \odot F_w(W_k) \right. \\
 & \quad \left. - |P_{k+1} - Q_k| \odot G_w(W_k) - ((P^*(W_k, Q_k) - Q_k) \odot F_w(W_k) - |P^*(W_k, Q_k) - Q_k| \odot G_w(W_k)) \right\|_{M_w(W_k)^\dagger}^2 \\
 & \leq \frac{\beta}{2\gamma q_{\max}} \|P^*(W_k, Q_k) - Q_k\|_{M_w(W_k)}^2 + \frac{2\beta q_{\max}^3}{\gamma} \|P_{k+1} - P^*(W_k, Q_k)\|_{M_w(W_k)^\dagger}^2.
 \end{aligned} \tag{93}$$

Substituting (91) and (93) into (90), we bound the second term in the RHS of (89) by

$$\begin{aligned}
 & -\frac{2}{\gamma} \mathbb{E}_{b'_k} [\langle P^*(W_k, Q_k) - Q_k, W_{k+1} - W_k \rangle] \\
 & \leq -\frac{\beta}{2\gamma q_{\max}} \|P^*(W_k, Q_k) - Q_k\|_{M_w(W_k)}^2 - \frac{\beta}{\gamma q_{\max}} \|(P^*(W_k, Q_k) - Q_k) \odot F_w(W_k) - |P^*(W_k, Q_k) - Q_k| \odot G_w(W_k)\|^2
 \end{aligned} \tag{94}$$

$$+ \frac{2\beta q_{\max}^3}{\gamma} \|P_{k+1} - P^*(W_k, Q_k)\|_{M_w(W_k)}^2.$$

The third term in the RHS of (89) follows the Lipschitz continuity of the analog update (Lemma A.2)

$$\begin{aligned} \frac{1}{\gamma^2} \mathbb{E}_{b'_k} [\|W_{k+1} - W_k\|^2] &= \frac{\beta^2}{\gamma^2} \|(P_{k+1} - Q_k) \odot F_w(W_k) - |P_{k+1} - Q_k| \odot G_w(W_k)\|^2 + \Theta\left(\frac{\beta \Delta w_{\min}}{\gamma^2}\right) \\ &\leq \frac{2\beta^2}{\gamma^2} \|(P^*(W_k, Q_k) - Q_k) \odot F_w(W_k) - |P^*(W_k, Q_k) - Q_k| \odot G_w(W_k)\|^2 \\ &\quad + \frac{2\beta^2}{\gamma^2} \|(P_{k+1} - Q_k) \odot F_w(W_k) - |P_{k+1} - Q_k| \odot G_w(W_k) \\ &\quad - ((P^*(W_k, Q_k) - Q_k) \odot F_w(W_k) - |P^*(W_k, Q_k) - Q_k| \odot G_w(W_k))\|^2 + \Theta\left(\frac{\beta \Delta w_{\min}}{\gamma^2}\right) \\ &\leq \frac{2\beta^2}{\gamma^2} \|(P^*(W_k, Q_k) - Q_k) \odot F_w(W_k) - |P^*(W_k, Q_k) - Q_k| \odot G_w(W_k)\|^2 + \frac{2\beta^2}{\gamma^2} \|P_{k+1} - P^*(W_k, Q_k)\|^2 \\ &\quad + \Theta\left(\frac{\beta \Delta w_{\min}}{\gamma^2}\right) \end{aligned} \quad (95)$$

Plugging (94) and (95) into (89) yields

$$\begin{aligned} \mathbb{E}_{b'_k} [\|P^*(W_{k+1}, Q_{k+1}) - Q_{k+1}\|^2] &\leq \|P^*(W_k, Q_k) - Q_k\|^2 - \frac{\beta}{2\gamma q_{\max}} \|P^*(W_k, Q_k) - Q_k\|_{M_w(W_k)}^2 + \Theta\left(\frac{\beta \Delta w_{\min}}{\gamma^2}\right) \\ &\quad - \left(\frac{\beta}{\gamma q_{\max}} - \frac{2\beta^2}{\gamma^2}\right) \|(P^*(W_k, Q_k) - Q_k) \odot F_w(W_k) - |P^*(W_k, Q_k) - Q_k| \odot G_w(W_k)\|^2 \\ &\quad + \frac{2\beta q_{\max}^3}{\gamma} \|P_{k+1} - P^*(W_k, Q_k)\|_{M_w(W_k)}^2 + \frac{2\beta^2}{\gamma^2} \|P_{k+1} - P^*(W_k, Q_k)\|^2. \end{aligned} \quad (96)$$

Notice the learning rate β is chosen as $\beta \leq \frac{\gamma}{2q_{\max}}$, we have

$$\begin{aligned} \mathbb{E}_{b'_k} [\|P^*(W_{k+1}, Q_{k+1}) - Q_{k+1}\|^2] &\leq \|P^*(W_k, Q_k) - Q_k\|^2 - \frac{\beta}{2\gamma q_{\max}} \|P^*(W_k, Q_k) - Q_k\|_{M_w(W_k)}^2 + \Theta\left(\frac{\Delta w_{\min}}{\gamma q_{\max}}\right) \\ &\quad - \frac{\beta}{2\gamma q_{\max}} \|(P^*(W_k, Q_k) - Q_k) \odot F_w(W_k) - |P^*(W_k, Q_k) - Q_k| \odot G_w(W_k)\|^2 \\ &\quad + \frac{2\beta q_{\max}^3}{\gamma} \|P_{k+1} - P^*(W_k, Q_k)\|_{M_w(W_k)}^2 + \frac{2\beta^2}{\gamma^2} \|P_{k+1} - P^*(W_k, Q_k)\|^2 \end{aligned} \quad (97)$$

which completes the proof. $\square$

E.4. Proof of Lemma E.3: Descent of accumulated asymmetric sequence $\varphi(P_k)$

In this section, we formally define the accumulated asymmetric function $\varphi(P) : \mathbb{R}^D \rightarrow \mathbb{R}^D$ element-wise defined by

$$[\varphi(P)]_d := \int_{\tau_i^{\min}}^{[P]_d} [G_p(P)]_d \, d[P]_d. \quad (98)$$

Lemma E.3 (Descent lemma of accumulated asymmetric sequence). *Under Assumption 2.1 and 3.6, it holds that*

$$\begin{aligned} \mathbb{E}_{\xi_k, b_k} [\varphi(P_{k+1})] &\leq \varphi(P_k) - \frac{\alpha C_\star}{2} \|G_p(P_k)\|^2 + \alpha^2 L q_{\max}^2 \sigma^2 + \frac{\alpha q_{\max}^2}{2C_\star} \|\nabla f(\bar{W}_k)\|^2 + \Theta\left(\frac{\Delta w_{\min}}{\gamma q_{\max}}\right) \\ &\quad + L \mathbb{E}_{\xi_k, b_k} \left[\|P_{k+1} - P_k + \alpha(\nabla f(\bar{W}_k; \xi_k) - \nabla f(\bar{W}_k)) \odot F_p(P_k) - b_k\|^2 \right]. \end{aligned} \quad (54)$$

Lemma E.3. The Lipschitz continuity of the generic response function ensures that the accumulated asymmetric function $\varphi(P)$ is L -smooth.

$$\mathbb{E}_{\xi_k, b_k} [\varphi(P_{k+1})] \leq \varphi(P_k) + \underbrace{\mathbb{E}_{\xi_k, b_k} [\langle G_p(P_k), P_{k+1} - P_k \rangle]}_{(a)} + \underbrace{\frac{L}{2} \mathbb{E}_{\xi_k, b_k} [\|P_{k+1} - P_k\|^2]}_{(b)} \quad (99)$$

Next, we will handle each term in the RHS of (99) separately.

Bound of the second term (a).

$$\begin{aligned} & \mathbb{E}_{\xi_k, b_k} [\langle G_p(P_k), P_{k+1} - P_k \rangle] \\ &= \mathbb{E}_{\xi_k} [\langle G_p(P_k), -\alpha \nabla f(\bar{W}_k, \xi_k) \odot F_p(P_k) - \alpha |\nabla f(\bar{W}_k, \xi_k)| \odot G_p(P_k) \rangle] \\ &= \langle G_p(P_k), -\alpha \nabla f(\bar{W}_k) \odot F_p(P_k) - \alpha \mathbb{E}_{\xi_k} [|\nabla f(\bar{W}_k, \xi_k)|] \odot G_p(P_k) \rangle \\ &= \left\langle \sqrt{\alpha \mathbb{E}_{\xi_k} [|\nabla f(\bar{W}_k, \xi_k)|]} \odot G_p(P_k), -\frac{\sqrt{\alpha}}{\sqrt{\mathbb{E}_{\xi_k} [|\nabla f(\bar{W}_k, \xi_k)|]}} \nabla f(\bar{W}_k) \odot F_p(P_k) - \sqrt{\alpha \mathbb{E}_{\xi_k} [|\nabla f(\bar{W}_k, \xi_k)|]} \odot G_p(P_k) \right\rangle \\ &\stackrel{(a)}{=} -\frac{1}{2} \left\| \sqrt{\alpha \mathbb{E}_{\xi_k} [|\nabla f(\bar{W}_k, \xi_k)|]} \odot G_p(P_k) \right\|^2 + \frac{1}{2} \left\| \frac{\sqrt{\alpha}}{\sqrt{\mathbb{E}_{\xi_k} [|\nabla f(\bar{W}_k, \xi_k)|]}} \nabla f(\bar{W}_k) \odot F_p(P_k) \right\|^2 \\ &\quad - \frac{1}{2} \left\| \frac{\sqrt{\alpha}}{\sqrt{\mathbb{E}_{\xi_k} [|\nabla f(\bar{W}_k, \xi_k)|]}} \nabla f(\bar{W}_k) \odot F_p(P_k) + \sqrt{\alpha \mathbb{E}_{\xi_k} [|\nabla f(\bar{W}_k, \xi_k)|]} \odot G_p(P_k) \right\|^2 \\ &\leq -\frac{\alpha}{2} \sum_{d \in [D]} \mathbb{E}_{\xi_k} [|\nabla f(\bar{W}_k, \xi_k)|_d] [G_p(P_k)]_d^2 + \frac{\alpha}{2} \sum_{d \in [D]} \frac{[\nabla f(\bar{W}_k)]_d^2}{\mathbb{E}_{\xi_k} [|\nabla f(\bar{W}_k, \xi_k)|_d]} [F_p(P_k)]_d^2 \\ &\stackrel{(b)}{\leq} -\frac{\alpha C_\star}{2} \|G_p(P_k)\|^2 + \frac{\alpha}{2C_\star} \|\nabla f(\bar{W}_k) \odot F_p(P_k)\|^2 \\ &\leq -\frac{\alpha C_\star}{2} \|G_p(P_k)\|^2 + \frac{\alpha q_{\max}^2}{2C_\star} \|\nabla f(\bar{W}_k)\|^2. \end{aligned} \quad (100)$$

The equality (a) holds by $\langle A, -B \rangle = -\frac{\|A\|^2}{2} - \frac{\|B\|^2}{2} + \frac{\|A-B\|^2}{2}$, the equality (b) holds by Assumption 3.6.

Bound the third term (b). Follow (83), we have:

$$\begin{aligned} & \frac{L}{2} \mathbb{E}_{\xi_k, b_k} [\|P_{k+1} - P_k\|^2] \\ &\leq L \mathbb{E}_{\xi_k, b_k} \left[\|P_{k+1} - P_k + \alpha (\nabla f(\bar{W}_k; \xi_k) - \nabla f(\bar{W}_k)) \odot F_p(P_k) - b_k\|^2 \right] + \alpha^2 L q_{\max}^2 \sigma^2 + \Theta(\alpha L \Delta w_{\min}). \end{aligned} \quad (101)$$

Substituting (100) and (101) into (99) and note that $\alpha \leq \frac{1}{2\gamma L q_{\max}}$, we have:

$$\begin{aligned} \mathbb{E}_{\xi_k, b_k} [\varphi(P_{k+1})] &\leq \varphi(P_k) - \frac{\alpha C_\star}{2} \|G_p(P_k)\|^2 + \alpha^2 L q_{\max}^2 \sigma^2 + \frac{\alpha q_{\max}^2}{2C_\star} \|\nabla f(\bar{W}_k)\|^2 + \Theta\left(\frac{\Delta w_{\min}}{\gamma q_{\max}}\right) \\ &\quad + L \mathbb{E}_{\xi_k, b_k} \left[\|P_{k+1} - P_k + \alpha (\nabla f(\bar{W}_k; \xi_k) - \nabla f(\bar{W}_k)) \odot F_p(P_k) - b_k\|^2 \right]. \end{aligned} \quad (102)$$

□

F. Additional Experiments and Experimental Setups

F.1. Device implementation details

In our experiments, we use two ReRAM array device presets from IBM AIHWKit simulator (Rasch et al., 2021): the HfO₂-based RRAM preset and the ReRamArrayOM preset in (Gong et al., 2022b). Both presets inherit from the SoftBoundsReferenceDevice model with potentiation and depression response functions as

$$q_+(w) = \alpha_+ \left(1 - \frac{w}{\tau_{\max}} \right), \quad q_-(w) = \alpha_- \left(1 + \frac{w}{\tau_{\min}} \right), \quad (103)$$

Table 3. Summary of the device parameters for the two HfO₂-based ReRAM models.

Device model	$(\tau_{\min}, \tau_{\max})$	$\Delta w_{\min}$	d2d variation $\sigma_{\pm}$	c2c variation σ_{c2c}
HfO ₂ -based ReRAM model (Gong et al., 2022b)	$(-1.0, 1.0)$	0.4622	0.7125	0.2174
ReRamArrayOMPresetDevice (Gong et al., 2022b)	$(-1.0, 1.0)$	0.0949	0.7829	0.4158

where $\alpha_+, \alpha_- > 0$ determine the effective update magnitudes for potentiation and depression, respectively. Following the SoftBounds-reference parameterization, we decompose these slopes as

$$\alpha_+ = \gamma + \rho, \quad \alpha_- = \gamma - \rho, \quad (104)$$

where γ controls the common slope magnitude and ρ controls the up/down asymmetry. In our model, these quantities are sampled independently for each cross-point as

$$\gamma_{ij} = \exp\left(\sigma_{d2d}\xi_{ij}^{(\gamma)}\right), \quad \rho_{ij} = \sigma_{\pm}\xi_{ij}^{(\rho)}, \quad (105)$$

where $\xi_{ij}^{(\gamma)}, \xi_{ij}^{(\rho)} \sim \mathcal{N}(0, 1)$. Thus, σ_{d2d} controls device-to-device variation in the slope magnitude, while $\sigma_{\pm}$ controls device-to-device variation in the potentiation/depression asymmetry.

At the array level, the response functions are applied elementwise. For a weight matrix W , we define

$$[Q_+(W)]_{ij} = q_{+,ij}(W_{ij}) = \alpha_{+,ij} \left(1 - \frac{W_{ij}}{\tau_{\max}}\right), \quad (106)$$

$$[Q_-(W)]_{ij} = q_{-,ij}(W_{ij}) = \alpha_{-,ij} \left(1 + \frac{W_{ij}}{\tau_{\min}}\right) \quad (107)$$

where each cross-point (i, j) has its own sampled device parameters (γ_{ij}, ρ_{ij}) .

Besides, IBM AIHWKit simulator (Rasch et al., 2021) also provides the cycle-to-cycle variation, which is modeled separately as a fresh random perturbation at each update pulse. For example, the scalar update response at cycle k can be written as

$$\Delta w_{ij,k}^+ = \Delta w_{\min} q_{+,ij}(w_{ij,k}) \left(1 + \sigma_{c2c}\zeta_{ij,k}^+\right), \quad (108)$$

$$\Delta w_{ij,k}^- = -\Delta w_{\min} q_{-,ij}(w_{ij,k}) \left(1 + \sigma_{c2c}\zeta_{ij,k}^-\right), \quad (109)$$

where $\zeta_{ij,k}^+, \zeta_{ij,k}^- \sim \mathcal{N}(0, 1)$ are sampled independently across devices and update cycles.

Under this model, there exists a unique SP at which the average positive and negative update sizes become equal; this weight is the SP $w^\diamond$. The ground truth value for SP of the device can be computed as:

$$w^\diamond = \frac{\alpha_+ - \alpha_-}{\frac{\alpha_+}{\tau_{\max}} - \frac{\alpha_-}{\tau_{\min}}} = \frac{2\rho}{\frac{\gamma+\rho}{\tau_{\max}} - \frac{\gamma-\rho}{\tau_{\min}}} \quad (110)$$

We evaluate two HfO₂-based ReRAM device models, whose non-idealities include device-to-device (d2d) and cycle-to-cycle (c2c) variation, update asymmetry, and limited conductance states, as summarized in Table 3.

F.2. Details of implementation for Figure 1 and Figure 2

In practice, however, the SP of an analog device is usually determined experimentally by applying alternating positive and negative update pulses until the weight converges. We denote by $r(N)$ the SP estimated in this way after N alternating pulses. To study how well this procedure recovers the true SP, we run simulations on the same 512×512 array with different pulse budgets $N \in \{500, 1000, 2000, 4000, 8000\}$, using a SoftBounds-based RPU preset with 2000 states. Figure 1a shows the empirical offsets of the SP mean and standard deviation over the 512×512 array, defined as the ground-truth statistics minus the corresponding estimated statistics. Here, the estimated SP statistics are computed by first estimating the SP for each array element, and then taking the mean and standard deviation across the 512×512 per-element SP estimates.

Table 4. Hyperparameters for TT-v2, AGAD, and E-RIDER used in the analog training runs for Table 1.

Hyperparameter	TT-v2	AGAD	E-RIDER
Units in mini-batch	\	True	True
Transfer frequency (transfer_every)	1.0	1.0	1.0
Transfer columns (transfer_columns)	True	True	True
Self-transfer (no_self_transfer)	True	True	True
Reads per transfer (n_reads_per_transfer)	1	1	1
Input chopping (in_chop_prob)	\	0.1	0.05
Input chopper random (in_chop_random)	\	False	False
Output chopping (out_chop_prob)	\	0.0	0.0
Fast learning rate (fast_lr)	0.005	0.01	0.5
Scale fast LR (scale_fast_lr)	\	False	False
Transfer learning rate (transfer_lr)	0.005	0.2	0.05
Scale transfer LR (scale_transfer_lr)	True	True	False
Auto granularity (auto_granularity)	\	1000	1000
Auto scale (auto_scale)	\	False	False
Auto momentum (auto_momentum)	\	0.99	0.99
Momentum (momentum)	0.1	0.0	0.0
Forget buffer (forget_buffer)	True	True	True
Tail weighting (tail_weightening)	\	5.0	5.0
Threshold scale (thres_scale)	0.8	\	\
Gamma (gamma)	\	\	0.1

As N increases, the distribution of $r(N)$ gradually moves towards and eventually almost coincides with the ground-truth distribution of $w^\diamond$. For smaller pulse counts, however, both the mean and the standard deviation of r can deviate significantly from those of $w^\diamond$. We model this mismatch element-wise using the following stochastic offset model:

$$r = w^\diamond + \mu_r + \sigma_r \xi, \quad \xi \sim \mathcal{N}(0, 1), \quad (111)$$

where μ_r captures the systematic offsets between the means of the two distributions and σ_r accounts for the variance offsets introduced by using a finite number of pulses. The combined term $o_r = \mu_r + \sigma_r \xi$ represents the residual offset on the reference device after SP subtraction. Our experiments in Figures 1a demonstrate that when the number of alternating pulses is not sufficiently large, this offset o_r introduces a non-negligible error in the estimated SP and the subsequent training. This effect is clearly visible in Figure 2, where we train a LeNet-5(MNIST) with TT-v1 using SPs estimated from different numbers of pulses: smaller N leads to significant degradation and even failure to converge.

Furthermore, the pulse cost required for accurate SP estimation increases as the device update granularity $\Delta w_{\min}$ improves. In Figure 1b, we sweep $\Delta w_{\min}$ from 5×10^{-3} down to 1.6×10^{-6} and, for each setting, estimate the SP using alternating pulses with a discrete pulse-budget schedule $N \in \{200, 500, 10^3, 2 \times 10^3, \dots, 8.192 \times 10^6\}$. We then report the smallest N such that the relative error of the estimated SP mean is within 1% of the ground-truth mean. The results show that higher-precision devices require substantially more pulses to reach the same accuracy target. Consequently, determining the SP by alternating pulses exhibits an inherent trade-off between pulse overhead and accuracy.

E.3. Hyperparameter settings for Tables 1 and 2

We report the hyperparameter settings used in our MNIST experiments with fully analog LeNet-5 and FCN models and the IO/analog readout noise configuration used in both MNIST and CIFAR 100 tasks. We use a fully analog LeNet-5-style CNN with two 5×5 convolution layers with 16 and 32 channels and two analog fully connected layers of sizes 512 and 128. The network uses tanh activations in the hidden layers. For the fully connected network, we set the input dimension of 784, two hidden layers of sizes 256 and 128, and a 10-class output layer. The model uses sigmoid activations in the hidden layers. For LeNet-5, we use a batch size of 8, while for the FCN we use a batch size of 10.

We focus our hyperparameter search on the learning-rate-related parameters, auto-granularity (threshold scale), and the input chopping probability. Consequently, the main distinctions between TT-v2, AGAD and E-RIDER are reflected in these tuned parameters. In addition, E-RIDER includes an extra residual scaling parameter γ , which controls the contribution of the residual term during gradient computation and is absent in AGAD. For the global learning rate on the LeNet model, we

Table 5. Hyperparameters for baseline algorithms and E-RIDER used in the CIFAR-100 fine-tuning experiments.

Category	TT-v2	AGAD	E-RIDER
Global learning rate (lr)	0.15	0.15	0.2
Fast learning rate (fast_lr)	0.01	0.01	0.1
Transfer learning rate (transfer_lr)	0.5	0.5	0.01
Scale transfer LR (scale_transfer_lr)	True	True	True
Momentum (momentum)	0.1	0.0	0.0
Threshold scale (thres_scale)	1.0	\	\
Input chopping (in_chop_prob)	\	0.1	0.05
Output chopping (out_chop_prob)	\	0.0	0.0
Auto granularity (auto_granularity)	\	1000	1000
Gamma (gamma)	\	\	0.1

Table 6. Hyperparameters for TT-v2, AGAD, and E-RIDER used in the analog training runs for Table 2.

Category	TT-v2	AGAD	E-RIDER
Input chopping (in_chop_prob)	\	0.1	0.05
Input chopper random (in_chop_random)	\	False	False
Output chopping (out_chop_prob)	\	0.0	0.0
Fast learning rate (fast_lr)	0.03	0.05	0.5
Scale fast LR (scale_fast_lr)	\	False	False
Transfer learning rate (transfer_lr)	0.01	0.3	0.05
Scale transfer LR (scale_transfer_lr)	True	True	False
Threshold scale (thres_scale)	0.8	\	\
Gamma (gamma)	\	\	0.1

set TT-v2 to 0.005, and AGAD and E-RIDER to 0.05. Other parameters are kept identical across methods unless stated otherwise. It is worth noting that we use a very small learning rate for TT-v2, because with a small number of states and a large reference-point offset, training becomes unstable and can diverge after several epochs when a larger learning rate is used. Despite the smaller learning rate, the accuracy we report for TT-v2 is measured after training has converged. The complete set of hyperparameter is reported in Table 4 for Lenet-5 and Table 6 for FCN.

Unless otherwise stated, TT-v2, AGAD and E-RIDER share the same IO and readout noise settings across both architectures: the input and output quantization resolutions are set to 7-bit and 9-bit, respectively; the output read noise is set to 0.06; and the input/output bounds are set to 1 and 12. This configuration is intended to reflect realistic analog hardware non-idealities; the corresponding parameters are summarized in Table 7.

F.4. Hyperparameter settings for Figure 4

For the ablation study comparing the total pulse budget of E-RIDER against the baseline zero-shifting algorithm, both methods use the same SoftBounds-based preset in the RPU configuration, and we set the desired update pulse length to be 5. For E-RIDER, we set the global learning rate to 0.1; the device-side learning rates are `fast_lr`=0.1 and `transfer_lr`=1, with `auto_granularity`=1000 and `in_chop_prob`=0.2. For the baseline zero-shifting setup, we use the TT-v2 algorithm after SP estimation is completed, train with a global learning rate of 0.05, and set `fast_lr`=0.01 and `transfer_lr`=1. We report the hyperparameter settings used in our CIFAR-100 experiments with ResNet-18. We replace the final fully connected layer and the last residual block with analog counterparts. For E-RIDER and AGAD, we report results after 80 epochs; for TT-v2, which consistently diverges during training, we report the best accuracy achieved before divergence. We use a batch size of 128, and decay the learning rate by a factor of 0.1 every 35 epochs. The complete set of hyperparameter is reported in Table 5.

F.5. Larger-scale ImageNet-1K fine-tuning experiment

To test the performance of E-RIDER on larger-scale dataset, we additionally evaluated E-RIDER on a reduced-scale ImageNet-1K setting. We construct a class-balanced training subset by sampling 200 images per class from the ImageNet-1K training split, and retain the full 50k-image validation set for evaluation. The architecture is a standard VGG-11-BN

Table 7. IO and analog readout noise settings for the forward, backward, and transfer-forward passes.

Parameter	Forward	Backward	Transfer-forward
MV type	ONE_PASS	ONE_PASS	ONE_PASS
Noise management	ABS_MAX	ABS_MAX	NONE
Bound management	ITERATIVE	ITERATIVE	NONE
Input bound (inp_bound)	1.0	1.0	1.0
Input resolution (inp_res)	0.0079365	0.0079365	0.0079365
Input noise (inp_noise)	0.0	0.0	0.0
Output bound (out_bound)	12.0	12.0	12.0
Output resolution (out_res)	0.0019608	0.0019608	0.0019608
Output noise (out_noise)	0.06	0.06	0.06
Input stochastic rounding	False	False	False
Output stochastic rounding	False	False	False

Table 8. Top-1 test accuracy for two dynamic SP tracking methods E-RIDER and AGAD on the reduced-scale ImageNet-1K setting under different reference mean/std. Best results are highlighted in bold.

Method	Mean Std	0.05	0.4	0.7	1.0
AGAD	0.05	63.82	63.40	63.60	63.67
E-RIDER		66.41	66.48	66.62	66.65
AGAD	0.20	63.61	63.77	64.06	64.05
E-RIDER		66.56	66.64	66.66	66.32
AGAD	0.30	63.74	64.05	63.75	64.09
E-RIDER		66.51	66.35	66.21	66.37
AGAD	0.40	64.02	63.66	63.64	64.07
E-RIDER		66.35	66.34	66.19	66.13

model with 8 convolutional layers and 3 fully connected layers, containing about 132M parameters. We replace the final two fully connected layers, fc2 and fc3, with analog layers of dimensions 4096×4096 and 4096×1000 , respectively.

For the non-ideal setting, we use the same non-perfect IO configuration as in Table 7. The hardware device we use is the ReRamArrayOMPresetDevice (Gong et al., 2022b). Its device-to-device variation, cycle-to-cycle variation, asymmetry, and limited conductance states are summarized in Table 3. The update asymmetry is modeled through soft bounds with separate up and down update responses provided in IBM AIHWKit simulator (Rasch et al., 2021).

The pretrained VGG-11-BN model achieves 70.18% top-1 test accuracy under digital evaluation. Direct deployment with non-ideal analog layers reduces test accuracy to 63.17%. We then fine-tune the analog model and compare E-RIDER against the dynamic SP-tracking baseline AGAD. Table 8 reports the resulting top-1 test accuracy under different offsets of the initial reference mean and standard deviation.

E-RIDER consistently improves over AGAD by 2 – 3% across all tested reference offsets. Both methods remain robust to imperfect initial SP estimation, but E-RIDER maintains a clear improvement. These results indicate that the proposed E-RIDER scales beyond the MNIST and CIFAR benchmarks used in the main paper.

F.6. Ablation studies of key hyperparameters

We provide additional ablation studies for three key E-RIDER hyperparameters: chopping probability p , moving averaging stepsize η and residual learning perturbation γ . First, Figure 5 shows the effect of the chopper probability p on MNIST-FCN. When $p = 0$, E-RIDER reduces to RIDER and the test accuracy drops sharply. For all tested $p > 0$, however, the final accuracy remains stable. We therefore use $p = 0.1$ as the default setting.

The moving-average stepsize η controls the update speed of the SP-tracking estimate. As shown in Table 9, performance is stable once $\eta > 0$. The best final accuracy occurs near our default $\eta = 0.5$.

The residual learning perturbation γ controls the contribution of the residual term in the gradient computation. Table 10 shows stable convergence for $\gamma \in [0.1, 0.4]$, with the best performance at $\gamma = 0.1$, which is set as the default value for E-RIDER. Starting at $\gamma = 0.5$, the final accuracy collapses.

Table 9. Ablation of the moving-average stepsize η in E-RIDER.

η	0.0	0.2	0.4	0.6	0.8	1.0
Final acc. (%)	89.43	91.46	93.57	92.06	91.38	92.01
Best acc. (%)	89.43	93.42	93.93	92.06	92.33	93.36

Table 10. Ablation of the residual learning perturbation γ in E-RIDER.

γ	0.1	0.2	0.3	0.4	0.5	0.6	0.7
Final acc. (%)	93.72	93.01	92.81	89.09	11.35	11.61	11.35
Best acc. (%)	93.72	93.01	92.81	89.09	33.04	56.88	49.61